NOTES

IDLE

Hello!!!

This note is mainly used to record some trivial but interesting problems that I encounter in daily life.

> *For as the heavens are higher than the earth,*
> *so are my ways higher than your ways,*
> *and my thoughts than your thoughts.*
>
> — Isaiah 55:9

Figure 1. Night City

Contents

[1] This section is organized by Ryan Ang and Qiutong Wang at Tianyuan Mathematics Research Centre

1. 2023.12.22

1.1. Identification of co-character groups with Lie algebras. Settings: G complex connected reductive algebraic group. $Q \subseteq P \subseteq \mathfrak{h}^*$ are root lattice and weight group, respectively. $\check{Q} \subseteq \check{P} \subseteq \mathfrak{h}$ are coroot lattice and coweight group, respectively. H is a maximal torus of G, $\mathfrak{h}$ is its Lie algebra.

There are identifications

$$\begin{array}{llll} I: & X_*(H) \otimes \mathbb{C} & \longrightarrow & \mathfrak{h} \\ & \phi & \longmapsto & d\phi(1) \end{array}$$

$$\begin{array}{llll} J: & X^*(H) \otimes \mathbb{C} & \longrightarrow & \mathfrak{h}^* \\ & \varphi & \longmapsto & d\varphi \end{array}$$

Under these identifications we have

$$\begin{aligned} \check{P} &= \{\check{\lambda} \in \mathfrak{h} \mid < \alpha, \check{\lambda} > \in \mathbb{Z}, \text{for all } \alpha \in Q\} \\ &= \{\check{\lambda} \in \mathfrak{h} \mid \exp(2\pi\sqrt{-1} < \alpha, \check{\lambda} >) = 1, \text{for all } \alpha \in Q\} \\ &= \{\check{\lambda} \in \mathfrak{h} \mid \alpha(\exp(2\pi\sqrt{-1}\check{\lambda})) = 1, \text{for all } \alpha \in Q\} \\ &= \{\check{\lambda} \in \mathfrak{h} \mid \exp(2\pi\sqrt{-1}\check{\lambda}) \in Z(G)\}. \end{aligned}$$

And

$$\begin{aligned} X_*(H) &= \{\check{\lambda} \in \mathfrak{h} \mid d\phi(\check{\lambda}) \in \mathbb{Z}, \text{for all } \phi \in X^*(H)\} \\ &= \{\check{\lambda} \in \mathfrak{h} \mid \exp(2\pi\sqrt{-1}d\phi(\check{\lambda})) = 1, \text{for all } \phi \in X^*(H)\} \\ &= \{\check{\lambda} \in \mathfrak{h} \mid \phi(\exp(2\pi\sqrt{-1}\check{\lambda})) = 1, \text{for all } \phi \in X^*(H)\} \\ &= \{\check{\lambda} \in \mathfrak{h} \mid \exp(2\pi\sqrt{-1}\check{\lambda}) = 1\}. \end{aligned}$$

Similarly

$$P = \{\lambda \in \mathfrak{h}^* \mid \exp(2\pi\sqrt{-1}\lambda) \in Z(\check{G})\}.$$

$$X^*(H) = \{\lambda \in \mathfrak{h}^* \mid \exp(2\pi\sqrt{-1}\lambda) = 1\}.$$

1.2. Pinnings of algebraic groups.

Theorem 1.1 (see AdC18)**.** *Inner automorphism group* Inn(G) *of G is equal to* G/Z(G)*,* Inn(G) *acts freely and transitively on the set of pinnings.*

I know that there is an analogy in compact connected groups, and it has been proved.

Theorem 1.2. *If* G *is a compact connected group, then its inner automorphism group* Inn(G) = G/Z(G) *acts freely and transitively on the set of pinnings. (where pinnings for it is a bit different from the complex case, that $(T, B_{\mathbb{C}}, \mathcal{X})$ is a pinning if T is a maximal torus, $B_{\mathbb{C}}$ a Borel of $G_{\mathbb{C}}$ containing T, $\mathcal{X}$ is a set of real rays in simple root space.)*

Proof. Here's proof from mathoverflow Lspise's answer:

To prove this, I'll use a few pieces of structure theory:

1. All maximal tori in G are G°-conjugate.
2. All Borel subgroups of $G_{\mathbb{C}}$ are $G^\circ_{\mathbb{C}}$-conjugate.
3. For every maximal torus T in G, the map $\mathrm{W}(G^\circ, T) \to \mathrm{W}(G^\circ_{\mathbb{C}}, T_{\mathbb{C}})$ is an isomorphism.
4. If $G_{\mathbf{sc}}$ and $(G_{\mathbb{C}})_{\mathbf{sc}}$ are the simply connected covers of the derived groups of G° and $G^\circ_{\mathbb{C}}$, then $(G_{\mathbf{sc}})_{\mathbb{C}}$ equals $(G_{\mathbb{C}})_{\mathbf{sc}}$.
5. Every compact Lie group has a finite subgroup that meets every component.

I only need (4) to prove that, for every maximal torus T in G, the map from T to the set of conjugation-fixed elements of $T/\mathrm{Z}(G^\circ)$ is surjective. This is probably a well known fact in its own right for real-group theorists.

Now consider triples $(T, B_{\mathbb{C}}, \mathcal{X})$ as follows: T is a maximal torus in G; $B_{\mathbb{C}}$ is a Borel subgroup of $G^\circ_{\mathbb{C}}$ containing $T_{\mathbb{C}}$, with a resulting set of simple roots $\Delta(B_{\mathbb{C}}, T_{\mathbb{C}})$; and $\mathcal{X}$ is a set consisting of a real ray in each complex simple root space (i.e., the set of positive real multiples of some fixed non-0 vector). (Sorry about the pair of modifiers "complex simple".) I will call these 'pinnings', although it doesn't agree with the usual terminology (where we pick individual root vectors, not rays). I claim that $G^\circ/\mathrm{Z}(G^\circ)$ acts simply transitively on the set of pinnings.

Once we have transitivity, freeness is clear: if $g \in G^\circ$ stabilises some pair $(T, B_\mathbb{C})$, then it lies in T, and so stabilises every complex root space; but then, for it to stabilise some choice of rays $\mathcal{X}$, it has to have the property that $\alpha(g)$ is positive and real for each simple root α; but also $\alpha(g)$ is a norm-1 complex number, hence trivial, for each simple root α, hence for each root α, so that g is central.

For transitivity, since (1) all maximal tori in G are G°-conjugate, so (2) for every maximal torus T in G, the Weyl group $\mathrm{W}(G^\circ_\mathbb{C}, T_\mathbb{C})$ acts transitively on the Borel subgroups of $G^\circ_\mathbb{C}$ containing $T_\mathbb{C}$, and (3) $\mathrm{W}(G^\circ, T) \to \mathrm{W}(G^\circ_\mathbb{C}, T_\mathbb{C})$ is an isomorphism, it suffices to show that all possible sets $\mathcal{X}$ are conjugate. Here's the argument that I came up with to show that they are even T-conjugate; I think it can probably be made much less awkward. Fix a simple root α, and two non-0 elements X_α and X'_α of the corresponding root space. Then there are a positive real number r and a norm-1 complex number z such that $X'_\alpha = rzX_\alpha$. Choose a norm-1 complex number w such that $w^2 = z$. There is then a unique element s_{ad} of $T_\mathbb{C}/\mathrm{Z}(G^\circ_\mathbb{C})$ such that $\alpha(s_{\mathrm{ad}}) = w$, and $\beta(s_{\mathrm{ad}}) = 1$ for all simple roots $\beta \ne \alpha$. By (4), we can choose a lift s_{sc} of s_{ad} to $(G_{\mathrm{sc}})_\mathbb{C} = (G_\mathbb{C})_{\mathrm{sc}}$, which necessarily lies in the preimage $(T_\mathbb{C})_{\mathrm{sc}}$ of (the intersection with the derived subgroup of) T, and put $t_{\mathrm{sc}} = s_{\mathrm{sc}} \cdot \overline{s_{\mathrm{sc}}}$. Then

$$\alpha(t_{\mathrm{sc}}) = \alpha(s_{\mathrm{sc}})\overline{\overline{\alpha}(s_{\mathrm{sc}})} = \alpha(s_{\mathrm{sc}})\overline{\alpha(s_{\mathrm{sc}})^{-1}} = w \cdot \overline{w^{-1}} = z,$$

and, similarly, $\beta(t_{\mathrm{sc}}) = 1$ for all simple roots $\beta \ne \alpha$. Now the image t of t_{sc} in $G^\circ_\mathbb{C}$ lies in $T_\mathbb{C}$ and is fixed by conjugation, hence lies in T; and $\mathrm{Ad}(t)X_\alpha = zX_\alpha$ lies on the ray through X'_α.

Since G also acts on the set of pinnings, we have a well defined map $p : G \to G^\circ/\mathrm{Z}(G^\circ)$ that restricts to the natural projection on G°. Now $\ker(p)$ meets every component, but it contains $\mathrm{Z}(G^\circ)$, so it need not be finite. Applying (5) to the Lie group $\ker(p)$ yields the desired subgroup H. Note that, as requested in your improved classification, conjugation by any element of H fixes a pinning, hence, if inner, must be trivial.

□

Remark 1.3. *The pinnings show that split reductive algebraic groups look like butterflies!*

According to Milne, Grothendieck used the following analogy to talk about the structure theory of algebraic groups. An algebraic group is like a butterfly. The body of the butterfly is the maximal torus. The wings are two opposite Borel subgroups. And the pins are the pinning maps.

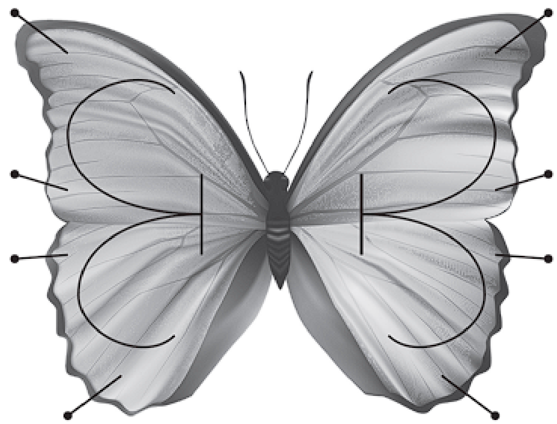


FIGURE 2. Pinning of butterfly

1.3. **Coherent continuation representations.** Until the end of today, G denotes a real reductive group in Harish-Chandra class.

Let's recall some basic definitions of coherent continuation representations, in order for me to do calculations in the case of real classical groups.

Fix a connected reductive complex Lie group $G_{\mathbb{C}}$, together with a Lie group homomorphism $\iota : \mathrm{G} \to \mathrm{G}_{\mathbb{C}}$ such that its differential $\mathrm{d}\iota : \mathrm{Lie}(\mathrm{G}) \to \mathrm{Lie}(\mathrm{G}_{\mathbb{C}})$ has the following two properties:

(1) the kernel of $\mathrm{d}\iota$ is contained in the center of $\mathrm{Lie}(\mathrm{G})$;
(2) the image of $\mathrm{d}\iota$ is a real form of $\mathrm{Lie}(\mathrm{G})$.

The analytical weight lattice of $\mathrm{G}_{\mathbb{C}}$ is identified with a subgroup of ${}^a\mathfrak{h}^*$ via $\mathrm{d}\iota$. We write $Q_\iota \subset {}^a\mathfrak{h}^*$ for this subgroup. Denote root lattice by $Q_{\mathfrak{g}}$, weight lattice by $Q^{\mathfrak{g}}$, then

$$Q_{\mathfrak{g}} \subset Q_\iota \subset Q^{\mathfrak{g}}.$$

They are all W stable subgroups. And we have some categories:

(1) $\mathrm{Rep}(\mathfrak{g}, Q_\iota)$ the category of all finite-dimensional representations of $\mathfrak{g}$ whose weight are contained in $Q_{\mathfrak{g}}$, it can be identified with $\mathrm{Rep}(\mathrm{Lie}(\mathrm{G}_{\mathbb{C}}), \mathrm{Q}_\iota)$ and hence $\mathrm{Rep}_{\mathrm{finite}}(\mathrm{G}_{\mathbb{C}})$;
(2) $\mathrm{Rep}(\mathrm{G})$ the category of all Casselman-Wallach representations of G, and its full subcategories $\mathrm{Rep}_\mu(\mathrm{G})$, $\mathrm{Rep}_{\mathrm{S}}(\mathrm{G})$.

To denote their Grothendieck groups, we replace Rep by $\mathcal{K}$.

Fix a Q_ι-coset $\Lambda = \lambda + Q_\iota \subset {}^a\mathfrak{h}^*$.

Definition 1.4 (coherent family)**.** *A $\mathcal{K}(G)$-valued Λ coherent family is a map*

$$\Phi : \Lambda \to \mathcal{K}(G)$$

satisfies the following two conditions:

(1) *for all $\nu \in \Lambda$, $\Phi(\nu) \in \mathcal{K}_\nu(G)$;*
(2) *for all representations $F \in \mathrm{Rep}(\mathfrak{g}, Q_\iota)$ and all $\nu \in \Lambda$,*

$$F \cdot (\Phi(\nu)) = \sum_{\mu} \Phi(\nu + \mu),$$

where μ runs over all weights of F, counted with multiplicities.

The set of coherent families is denoted by $\mathrm{Coh}_\Lambda(\mathcal{K}(G))$

We have the integral Weyl group $W_\Lambda = \{w \in W \mid w\lambda - \lambda \in Q_\iota\}$ there is a representation of W_Λ on $\mathrm{Coh}_\Lambda(\mathcal{K}(G))$ by

$$(w \cdot \Phi)(\nu) = \Phi(w^{-1}\nu).$$

This is called a coherent continuation representation.

2. 2023.12.23

2.1. **Review of regular parameter.** Following yesterday's discussion of coherent continuation representations, we define two basis for it, use Langland classification.

Suppose H is a Cartan subgroup of G, by which we mean a subgroup that is the centralizer of a Cartan subalgebra of $\mathrm{Lie}(G)$ (In the case of linear groups, it's just a real algebraic torus.). H has a unique maximal compact subgroup (since G is in Harish-Chandra's class, adjoint representation of H is trivial, H_0 lies in the center of H, as we know, all maximal subgroups of Lie groups are conjugate to each other, use Cartan decomposition of H, we win. In fact, H has a unique maximal compact torus and a unique maximal split torus) We denote by $\Delta_{\mathfrak{h}} \subset \mathfrak{h}^*$ the root system of $\mathfrak{g}$. A root is called imaginary if $\check{\alpha} \in \mathfrak{t}$ (i.e. roots which take pure imaginary values on $\mathrm{Lie}(H)$), an imaginary root α is called compact if $\mathfrak{g}_\alpha$ and $\mathfrak{g}_{-\alpha}$ is contained in the complexified Lie algebra of a common compact subgroup of G.(this is equivalent to $\mathfrak{g}_\alpha$ belong to a Lie subalgebra of a compact subgroup, since $\mathfrak{g}_\alpha$ lies in a Lie subalgebra of maximal compact subgroup is equivalent to $\mathfrak{g}_{-\alpha}$ Lies in the same subgroup)

There are two important facts about the representation theory of H:

(1) Every Casselman-Wallach representation of H is finite-dimensional, it follows directly from that H is an extension of an abelian group and a finite group;
(2) For every $\Gamma \in \mathrm{Irr(H)}$ differential of Γ is a direct sum of one-dimensional representations attached to a unique $\mathrm{d}\Gamma \in \mathfrak{h}^*$. (it is true because H is in Harish-Chandra's class).

For every Borel subalgebra $\mathfrak{b}$ of $\mathfrak{g}$ containing $\mathfrak{h}$, write

$$\xi_{\mathfrak{b}} : \mathfrak{h} \to {}^a\mathfrak{h},$$

for the linear isomorphism attached to $\mathfrak{b}$ defined by

$$\mathfrak{h} \hookrightarrow \mathfrak{b} \to \mathfrak{b}/[\mathfrak{b},\mathfrak{b}] = {}^a\mathfrak{h}.$$

The transpose inverse of this map is still denoted by $\xi_{\mathfrak{b}} : {}^a\mathfrak{h}^* \to \mathfrak{h}^*$.

Write

$$W({}^a\mathfrak{h}^*, \mathfrak{h}^*) := \{\xi_{\mathfrak{b}} : {}^a\mathfrak{h}^* \to \mathfrak{h}^* \mid \mathfrak{b} \text{ is a Borel subalgebra containing } \mathfrak{h}\}.$$

Definition 2.1. *For every element $\xi \in W({}^a\mathfrak{h}^*, \mathfrak{h}^*)$, put*

$$\delta(\xi) := \frac{1}{2} \cdot \sum_{\alpha \text{ is an imaginary root in } \xi\Delta^+} \alpha - \sum_{\beta \text{ is an compact imaginary root in } \xi\Delta^+} \beta \in \mathfrak{h}^*.$$

Write $\mathcal{P}_\Lambda(G)$ for the set of all triples $\gamma = (H, \xi, \Gamma)$, where H is a Cartan subgroup of G, $\xi \in W({}^a\mathfrak{h}^*, \mathfrak{h}^*)$, and

$$\begin{array}{rccc} \Gamma : & \Lambda & \longrightarrow & \mathrm{Irr(H)} \\ & \nu & \longmapsto & \Gamma_\nu \end{array}$$

is a map with the following properties:

(1) *$\Gamma_{\nu+\beta} = \Gamma_\nu \otimes \xi(\beta)$ for all $\beta \in Q_\iota$ and $\nu \in \Lambda$;*
(2) *$\mathrm{d}\Gamma_\nu = \xi(\nu) + \delta(\xi)$ for all $\nu \in \Lambda$.(note that Cartan subgroups not always have irr repn with this differential!)*

Here $\xi(\beta)$ is naturally viewed as a character of H by using the homomorphism $\iota : H \to H_{\mathbb{C}}$, and $H_{\mathbb{C}}$ is a Cartan subgroup of $G_{\mathbb{C}}$ containing $\iota(H)$.

2.2. **Outer automorphism of algebraic groups.**

Lemma 2.2 (a simple discover of abstract algebra)**.** *G an abstract group, H a normal subgroup of G, if G acts transitively on a set X such that the restriction of this action to H is free and transitive, then the short exact sequence of groups*

$$1 \to H \to G \to G/H \to 1$$

Proof. Actually choose an element $x \in X$, we have

$$G \simeq \mathrm{Stab}_G(x) \ltimes H.$$

□

Using this lemma, we can see that for a complex connected reductive algebraic group G, $\mathrm{Aut}(G)$ acts on the set of pinnings transitively, with its normal subgroup the $\mathrm{Inn}(G)$ acts freely and transitively, so every pinning gives us a split of the short exact sequence

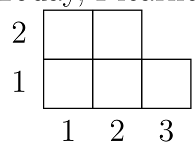
$$1 \to \mathrm{Inn}(G) \to \mathrm{Aut}(G) \to \mathrm{Out}(G) \to 1.$$

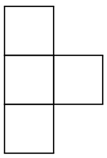
Some facts about split connected reductive algebraic groups over arbitrary field k:

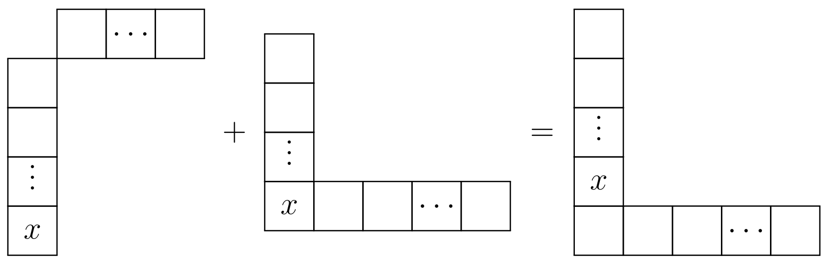
(1) $\{\text{split connected reductive algebraic groups}\}/_{\backsim} = \{\text{root datums}\}/_{\backsim}$;
(2) For any two automorphisms of G, if they induce the same automorphism of root datum, then they differ by an inner automorphism;
(3) Fix a pinning, we get an isomorphism $\mathrm{Out}(G) \simeq \mathrm{Aut}(D_b)$.

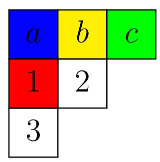
3. 2024.2.27

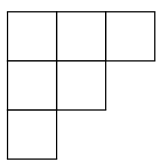
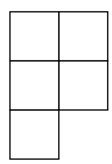
Today, I learned how to draw the Young tableau.

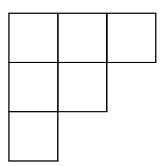


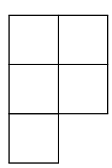


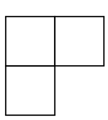


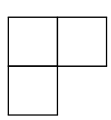


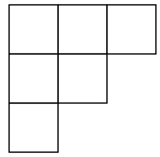
 

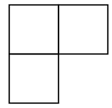




ce est chinois  english

$a =$  $+ b$

youngdiagram1: 

young2: 

For a compact connected Lie group G, choose a maximal torus T, there is a canonical 1-1 correspondence:

$$X^*(T)/W \longleftrightarrow (X^*(T)+\rho)/W \longleftrightarrow \mathrm{Irr}(G)$$

Moreover, if we choose a positive system Ψ^+, and then ρ_{Ψ^+}, the above sets also canonically isomorphic to the set of dominant analytically integral weights:

$$\{\lambda \in X^*(T) \,|\, \langle \lambda, \alpha \rangle \geq 0, \text{for all } \alpha \in \Psi^+\}$$

This can be generalized to parameter of discrete series representations for connected semisimple Lie groups G with compact Cartan subgroup T :

$$(X^*(T)+\rho)^{reg}/W_c$$

If a discrete series representation V has minimal K type which has maximal weight ν, then it's parameter is $\lambda = \nu - \rho + 2\rho_c$.(not sure)

Note that for a fixed infinitesimal character, there are exactly $|W/W_c|$ different discrete series representations.

4. 2024.2.28

Some facts about continuous complex characters of some common abelian topological groups:

field	character group	correspondence
$(\mathbb{R}_+^\times, \times)$	$\mathbb{C}$	$\xi \mapsto (t \mapsto t^\xi)$
$(\mathbb{C}^\times, \times)$	$\mathbb{C} \times \mathbb{Z}$	$(\xi, n) \mapsto (z \mapsto \|z\|^\xi (\frac{z}{\|z\|})^n)$
$(\mathbb{S}^1, \times)$	$\mathbb{Z}$	$n \mapsto (z \mapsto z^n)$
$(\mathbb{Q}_p, +)$	$\mathbb{Q}_p$	$\xi \mapsto (x \mapsto \exp(2\pi i \xi \{x\}_p))$

Remark 4.1. *Note that all continuous characters of $\mathbb{Q}_p$ are automatically locally constant (smooth), which is consistent with the case of real lie groups.*

5. 2024.10.22

Lemma 5.1. *The associated variety of an irreducible Casselman-Wallach representation of a real reductive group in Harish-Chandra class is the closure of a nilpotent orbit in* $\mathfrak{g}^*$.

Proof. By the work of Borho, Brylinski [BB82], and Joseph [Jos85], the associated variety of any primitive ideal (annihilator of an irreducible $\mathfrak{g}$-module) is the closure of a single nilpotent orbit. So we only need to prove that $\mathrm{gr}(\mathrm{Ann}_{\mathcal{U}(\mathfrak{g})}(\mathrm{V}))$ is a primitive ideal.

Since the K-finite part V_K is dense in V, $\mathrm{Ann}_{\mathcal{U}(\mathfrak{g})}(V) = \mathrm{Ann}_{\mathcal{U}(\mathfrak{g})}(V_K)$, we only need to consider the annihilator of the irreducible $(\mathfrak{g}, K)$-module V_K.

If G is connected, then K is also connected, so V_K is irreducible as $\mathcal{U}(\mathfrak{g})$-module, in this case, $\mathrm{Ann}_{\mathcal{U}(\mathfrak{g})}(V)$ is a primitive ideal.

If G is not connected, let G_0 be the identity component of G, $K_0 = G_0 \cap K$ is a maximal compact subgroup of G_0, then $V_K|_{G_0}$ is a finite length $(\mathfrak{g}, K_0)$-module, it has an irreducible quotient $V_K \to V_0$, then according to the theory of induction of $(\mathfrak{g}, K)$-modules [?, Chapter 2] we have an injective morphism of $(\mathfrak{g}, K_0)$-modules:

$$V_K|_{K_0} \hookrightarrow (\mathrm{induced}_{K_0}^{K}(V_0))|_{K_0} \cong \bigoplus_{\text{double cosets } K_0 k K_0} kV_0, \tag{5.1}$$

where kV_0 be the $(\mathfrak{g}, K_0)$-module whose underlying space is V_0, whose action by $X \in \mathfrak{g}$ is the usual action by $\mathrm{Ad}(k)X$, whose action by $g \in K_0$ is the usual action by $k^{-1}gk$, it has the same annihilator in $\mathcal{U}(\mathfrak{g})$ as V_0 since G is in Harish-Chandra's class, and hence the annihilator of V_K is the same as the annihilator of V_0, which is a primitive ideal.

$\square$

6. 2024.11.28

Any split connected reductive group over arbitrary field can be uniquely defined and split over $\mathbb{Z}$. (Chevalley group)

Classification of split reductive group is the same over arbitrary non-empty schemes, which is equivalent to the classification of root data.(cf. [SGA3])

7. 2024.12.17

Geometric Pre-quantization

Let X be a smooth manifold. There are natural maps

$$\mathrm{Pic}(X) \longrightarrow \mathrm{H}^2_{dR}(X;\mathbb{R}) \simeq \mathrm{H}^2_{Cech}(X;\mathbb{R}).$$

Construction of the first map: For a (complex) line bundle $\mathbb{L} \in \mathrm{Pic}^h(X)$, let $\langle,\rangle$ be a Hermitian metric on it and ∇ be a unitary connection. (Every line bundle $\mathbb{L} \in \mathrm{Pic}(X)$ has a Hermitian metric and a unitary connection on it.)

- choose a locally trivialization $\{U_\alpha\}$ of $\mathbb{L}$, such that all U_α and $U_\alpha \cap U_\beta$ are contractible;
- choose Unitary local frames $\{s_\alpha\}$ for $\mathbb{L}$ on each U_α, then $s \leftrightarrow (f_\alpha)$ and $\nabla \leftrightarrow (\theta_\alpha)$ (unitary implies that θ_α are pure imaginary);
- define $\Omega := \mathrm{d}\theta_\alpha$, and $c_1(\mathbb{L}) := [\frac{1}{2\pi i}\Omega] \in \mathrm{H}^2_{dR}(X;\mathbb{R})$ ($\Omega = \mathrm{curv}\nabla$).

$c_1(\mathbb{L})$ is independent of the choice of $\langle,\rangle$ and ∇ by Chern-Weil theory. And we can see that $\theta_\alpha - \theta_\beta = \frac{\mathrm{d}g_{\alpha,\beta}}{g_{\alpha,\beta}} = \mathrm{d}\log(g_{\alpha,\beta})$ on $U_\alpha \cap U_\beta$, where $g_{\alpha,\beta}$ is the translation function ($s_\beta = g_{\alpha,\beta}s_\alpha$).

Construction of the second map (classical de-Rham isomorphism), choose a 2-form $\omega \in \mathrm{H}^2_{dR}(X;\mathbb{R})$

- choose a cover $\{U_\alpha\}$ of $\mathbb{L}$, such that all U_α and $U_\alpha \cap U_\beta$ are contractible;
- on each $U_\alpha, choose a 1-form \theta_\alpha$ such that $\mathrm{d}\theta_\alpha = \omega|_{U_\alpha}$;
- since $\mathrm{d}\theta_\alpha = \mathrm{d}\theta_\beta$ on $U_\alpha \cap U_\beta$, we can choose smooth function $c_{\alpha,\beta}$ such that $\mathrm{d}c_{\alpha,\beta} = \theta_\alpha - \theta_\beta$;
- let $c_{\alpha,\beta,\gamma} = c_{\alpha,\beta} + c_{\beta,\gamma} + c_{\gamma,\alpha}$, which is a constant, and we get the corresponding 2-cocycle $(c_{\alpha,\beta,\gamma}) \in \mathrm{H}^2_{Cech}(X;\mathbb{R})$.

If ω come from the first map, we can choose $c_{\alpha,\beta} = \frac{1}{2\pi i}\log g_{\alpha,\beta}$, and hence $c_{\alpha,\beta,\gamma} = \frac{1}{2\pi i}(\log g_{\alpha,\beta} + \log g_{\beta,\gamma} + \log g_{\gamma,\alpha}) \in \mathbb{Z}$. So the morphism is in fact

$$\mathrm{Pic}(X) \longrightarrow \mathrm{H}^2_{dR}(X;\mathbb{Z}) \simeq \mathrm{H}^2_{Cech}(X;\mathbb{Z}).$$

In fact the first map is also an isomorphism. (cf. prequantization) For a 2-form ω, choose θ_α and $c_{\alpha,\beta}$ as above, let $g_{\alpha,\beta} = \exp(2\pi i c_{\alpha,\beta})$, these are transition functions of a complex line bundle $\mathbb{L}$.

Corollary 7.1. *Assume that G is a connected Lie group. The following are equivalent:*

- $\Omega \subseteq \mathfrak{g}^*$ *is integral* ($[\sigma_\Omega] \in \mathrm{H}^2(X;\mathbb{Z})$)*;*
- *there exists a G-equivariant complex line bundle over Ω with a G-invariant Hermitian connection ∇ such that*

$$\frac{1}{2\pi i}\mathrm{curv}(\nabla) = \sigma_\Omega.$$

Theorem 2.6 (Weil). *Let M be a smooth manifold and ω a real, closed 2-from whose cohomology class $[c]$ is integral. Then there is a unique Hermitian line bundle $\mathbb{L}$ over M with unitary connection ∇ so that $c_1(\mathbb{L}) = [c]$.*

Sketch of Proof. Existence: Reverse the arguments above. Since $[c]$ is integral, $e^{2\pi c_{ijk}} = 1$. As a consequence, $g_{ij}g_{jk}g_{kl} = 1$. So g_{ij}'s are the transition function for some line bundle whose first Chern class is $[c]$.

Uniqueness: Suppose $c_1(\mathbb{L}) = c_1(\widetilde{\mathbb{L}})$. Let $h_{ij} = \frac{1}{2\pi i} g_{ij}$ and define $\tilde{h}_{ij}$ similarly. Then the functions $\hat{h}_{ij} = h_{ij} - \tilde{h}_{ij}$ satisfies the relation

$$\hat{h}_{ij} + \hat{h}_{jk} + \hat{h}_{ki} = 0.$$

We take a partition of unity ρ_k and let $\lambda_i = e^{2\pi i \sum \hat{h}_{ki}\rho_k}$. Then

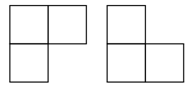
$$\lambda_i g_{ij} \lambda_j^{-1} = e^{2\pi i (h_{ij} + \sum(\hat{h}_{ki} - \hat{h}_{kj})\rho_k)} = e^{2\pi i (h_{ij} - \hat{h}_{ij})} = \tilde{g}_{ij}.$$

So as line bundles $\mathbb{L} \simeq \widetilde{\mathbb{L}}$. ∎

Figure 3. Proof

8. 2025.4.7

TooYoung package from xiongrui.



1	$i+1$
2	
3	
$\dots$	
i	
$i+2$	
$\dots$	
n	

9. 2025.8.14 [2]

Let G be a finite central extension of an open subgroup of $\mathbb{R}$-points of a connected reductive algebraic group (reductive Nash group [Sun15]) with Lie algebra $\mathfrak{g}$. Fix an invariant nondegenerate bilinear form $B(-,-)$ on $\mathfrak{g}$.

For an irreducible Casselman-Wallach representation π of G, let $\mathrm{Wh}_{\mathcal{O}}(\pi)$ be the space of generalized Whittaker models of π associated to the nilpotent orbit $\mathcal{O}$. Let

$$\mathrm{WO}(\pi) := \{\mathcal{O} \,|\, \mathcal{O} \text{ is a nilpotent orbit such that } \mathrm{Wh}_{\mathcal{O}}(\pi) \neq 0\},$$

and the *Whittaker support* $\mathrm{WS}(\pi)$ of π be the set of all maximal orbits in $\mathrm{WO}(\pi)$ with respect to closure inclusion:

$$\mathrm{WS}(\pi) := \{\mathcal{O} \,|\, \mathcal{O} \text{ is a maximal orbit in } \mathrm{WO}(\pi)\}.$$

Another nilpotent invariant is the wavefront cycle:

$$\mathcal{WF}(\pi) = \sum_i m_i[\mathcal{O}_i].$$

With the corresponding wavefront set :

$$\mathrm{WF}(\pi) = \bigcup_{m_i \neq 0} \overline{\mathcal{O}_i}.$$

Mœglin-Waldspurger [MW87] proved that $\mathrm{WF}(\pi) = \mathrm{WS}(\pi)$ when F is a p-adic field. We want to test this conjecture for arbitrary reductive almost linear Nash groups.

Conjecture 9.1 (Main conjecture)**.** *For any reductive almost linear Nash group G and any irreducible Casselman-Wallach representation π:*

$$\mathrm{WF}(\pi) = \mathrm{WS}(\pi).$$

Take a nilpotent orbit $\mathcal{O} \subseteq \mathfrak{g}$ (or $\mathfrak{sl}_2$-triple $\gamma = \{Y, H, X\}$ in $\mathfrak{g}$). $\mathfrak{g}/\mathfrak{g}^Y$ is a symplectic space with symplectic form $\langle -,- \rangle := B(Y, [-,-])$. Set $G^Y := \mathrm{Stab}_G(Y)$, there is a natural homomorphism $G^Y \to \mathrm{Sp}(\mathfrak{g}/\mathfrak{g}^Y, \langle -,- \rangle)$. Let $\widetilde{G^Y}$ be the double cover of G^Y induced by the metaplectic double cover of $\mathrm{Sp}(\mathfrak{g}/\mathfrak{g}^Y, \langle -,- \rangle)$.

Definition 9.2. *The nilpotent $\mathcal{O}$ is called admissible if the double cover $\widetilde{G^Y}$ is split over the identity component of $(G^Y)_0$. It is called quasi-admissible if $\widetilde{G^Y}$ has a finite-dimensional genuine representation.*

Result of [GGS21] says that $\mathrm{WS}(\pi)$ consists of quasi-admissible orbits. But they do not know whether the non-admissible quasi-admissible orbits appear in Whittaker supports of representations. We may ask the same question for the wavefront set:

Problem 1: Are all the nilpotent orbits in the wavefront cycle admissible?

For the case when G is general linear, it is well known that the wavefront set $\mathrm{WF}(\pi)$ of an irreducible Casselman-Wallach representation π of G consists of a single nilpotent adjoint orbit. Together with some counting results, it maybe possible to construct all irreducible representations of general linear group via theta correspondence and coherent continuation.

Problem 2: Can we construct all irreducible representations via theta correspondence, coherent continuation, and twist of characters?

---

[2] This section is organized by Ryan Ang and Qiutong Wang at Tianyuan Mathematics Research Centre

Assuming the construction steps above are done, it then remains to study the Whittaker support $\mathrm{WS}(\pi)$ of π. We focus on two types of interactions with Whittaker models - theta correspondence and coherent continuation.

On one hand, Gomez-Zhu [GZ14] proved for type I reductive (symplectic-orthogonal) dual pairs in the stable range that the big theta lift preserves the structure of Whittaker models:

$$\mathrm{Wh}_{\Theta(\mathcal{O})}\left(\Theta(\pi)\right) = \mathrm{Wh}_{\mathcal{O}}(\pi^{\vee}).$$

Noting that $(G, G') = (\mathrm{GL}_n(F), \mathrm{GL}_m(F))$ is a (reductive) dual pair for any $n, m \in \mathbb{N}$, one observes that if one can prove an analogue of Gomez-Zhu's result for type II dual pairs, one can use it to infer properties of the structure of the Whittaker support $\mathrm{WS}(\pi)$. Hence we propose the following subconjecture:

Conjecture 9.3. *If $(G, G') = (\mathrm{GL}_n(F), \mathrm{GL}_m(F))$ is a type II reductive dual pair over F in the stable range (with $n \leq m$), and π is a genuine Casselman-Wallach representation of G, then*

$$\mathrm{Wh}_{\Theta(\mathcal{O})}\left(\Theta(\pi)\right) = \mathrm{Wh}_{\mathcal{O}}(\pi^{\vee}),$$

where $\pi^{\vee}$ denotes the smooth contragredient of π. Furthermore, the above will imply that

$$\mathrm{WS}\left(\Theta(\pi)\right) = \mathrm{WS}(\pi^{\vee}).$$

On the other hand, Barbash-Vogan [BV80] proved that under coherent continuation, the multiplicity of wavefront cycle changes in the form of a Goldie rank polynomial, i.e.

$$\mathcal{WF}(\pi(\lambda)) = p(\lambda) \cdot (\sum_i c_i[\mathcal{O}_i]),$$

where $p(\lambda)$ is the goldie rank polynomial, λ is dominant. We expect some similar result for the multiplicity of the generalized Whittaker model.

Problem 3: For a fixed nilpotent orbit $\mathcal{O}$, does there exist a constant $c_{\mathcal{O}}$ such that $\mathrm{dim}\mathrm{Wh}_{\mathcal{O}}(\pi(\lambda)) = c_{\mathcal{O}} \cdot p(\lambda)$?

10. 2025.10.1

Orbit method for nilpotent Lie groups.

Let G be a connected, simply connected nilpotent Lie group with Lie algebra $\mathfrak{g}$. Let $\mathfrak{g}^*$ be the dual space of $\mathfrak{g}$.

We have fourier transform:

$$\mathcal{F} : \mathcal{S}(\mathfrak{g})\mathrm{dX} \to \mathcal{S}(\sqrt{-1}\mathfrak{g}^*), \quad f\mathrm{dX} \mapsto \widehat{f\mathrm{dX}}, \quad \widehat{f\mathrm{dx}}(l) = \int_{\mathfrak{g}} f(X)e^{\langle l,X\rangle}dX.$$

Take its dual:

$$\mathcal{S}'(\sqrt{-1}\mathfrak{g}^*) \to (\mathcal{S}(\mathfrak{g})\mathrm{dX})', \quad T \mapsto \check{T}, \quad \langle \check{T}, f\mathrm{dX}\rangle = \langle T, \widehat{f\mathrm{dX}}\rangle.$$

For any irreducible unitary representation π of G, we have its character $\Theta_\pi \in C^{-\infty}(G)$ defined by

$$\Theta_\pi(f) = \mathrm{Tr}(\pi(f\mathrm{dg})), \quad f\mathrm{dg} \in C_c^\infty(G)\mathrm{dg}.$$

Using the exponential map $\exp : \mathfrak{g} \to G$ (isomorphism in this case), we can view Θ_π as a distribution on $\mathfrak{g}$:

$$\theta_\pi(f\mathrm{dX}) = \Theta_\pi(f \circ \exp), \quad f\mathrm{dX} \in C_c^\infty(\mathfrak{g})\mathrm{dX}.$$

In fact, θ_π is a tempered generalized function on $\mathfrak{g}$ (it can continuously extends to $\mathcal{S}(\mathfrak{g})\mathrm{dX}$). By the work of Howe, Kirillov, and others, we have the following theorem:

Theorem 10.1. *There exists a unique coadjoint orbit $\mathcal{O} \subseteq \sqrt{-1}\mathfrak{g}^*$ such that $\check{\theta}_\pi = \mu_{\mathcal{O}}$, where $\mu_{\mathcal{O}}$ is the canonical measure on $\mathcal{O}$.*

This defines a bijection between $\widehat{G}$ (the unitary dual of G) and the set of coadjoint orbits in $\sqrt{-1}\mathfrak{g}^*$.

Conversely, for any coadjoint orbit $\mathcal{O} \subseteq \sqrt{-1}\mathfrak{g}^*$, we can construct an irreducible unitary representation $\pi_{\mathcal{O}}$ of G such that $\check{\theta}_{\pi_{\mathcal{O}}} = \mu_{\mathcal{O}}$. This is the more interesting part, use the method of geometric quantization.

Take any element $\lambda \in \mathcal{O}$, we have an identification $\mathcal{O} = G/G_\lambda$, $lambda : \mathfrak{g}_\lambda \to \mathbb{C}$ is a Lie algebra homomorphism, and hence $\chi_\lambda : G_\lambda \to \mathbb{C}^\times$ defined by $\chi_\lambda(\exp X) = e^{\langle\lambda,X\rangle}$ is a unitary character of G_λ. Then we can defined a Hermitian line bundle $\mathcal{L}_\lambda = G \times^{G_\lambda} \mathbb{C}$ over $\mathcal{O}$, with a natural G-invariant connection ∇ such that $\mathrm{curv}(\nabla) = \omega_{\mathcal{O}}$, where $\omega_{\mathcal{O}}$ is the Kirillov-Kostant-Souriau symplectic form on $\mathcal{O}$.

Then we can define the Hilbert space of square integrable sections of $\mathcal{L}_\lambda$, but this space is too large. We need to choose a polarization $\mathfrak{h} \subseteq \mathfrak{g}_{\mathbb{C}}$ (maximal isotropic subalgebra with respect to the bilinear form $(X,Y) \mapsto \langle\lambda,[X,Y]\rangle$) and consider the space of square integrable holomorphic sections of $\mathcal{L}_\lambda$ which are covariant constant along $\mathfrak{h}$.

In fact the polarization $\mathfrak{h}$ corresponds to a G-invariant integrable distribution on $\mathcal{O}$, its a subbundle P of $T\mathcal{O}$ each fiber is a lagrangian subspace. Then we can define the space of polarized sections:

$$\Gamma_P(\mathcal{O}, \mathcal{L}_\lambda) = \{s \in \Gamma^\infty(\mathcal{O}, \mathcal{L}_\lambda \otimes |\omega_{\mathcal{O}/\mathcal{P}}|^{\frac{1}{2}}) \mid \nabla_X s = 0, \forall X \in P\}.$$

On this space, we have a unitary G action and get a unitary representation $\pi_{\mathcal{O}}$ of G, it turns out that $\pi_{\mathcal{O}}$ is irreducible and independent of the choice of the polarization $\mathfrak{h}$.

Kirillov's philosophy works well for nilpotent Lie groups, but for more general Lie groups, it is not true anymore.

Now assume G is a real reductive group, with Lie algebra $\mathfrak{g}_0$. For any irreducible Casselman-Wallach representation π of G, we still have its generalized function character $\Theta_\pi \in C^{-\infty}(G)$, and hence a tempered generalized function θ_π on $\mathfrak{g}$. Here $\theta_\pi \in C^{-\infty}(\mathfrak{g}_0)$ is defined as follows:

$$\theta_\pi(f\mathrm{dX}) = \Theta_\pi(\exp_*(\sqrt{\det(\exp)}f\mathrm{dX})), \quad f\mathrm{dX} \in C_c^\infty(\mathfrak{g}_0)\mathrm{dX}.$$

The generalized function Θ_π has the following properties:

- Θ_π is invariant under conjugation of G;
- Θ_π is an eigendistribution of $\mathcal{U}(\mathfrak{g})^G$;
- Θ_π is locally integrable, and real analytic on the regular semisimple locus G^{rs}.

By the work of Rossmann, we have the following theorem:

Theorem 10.2. *If π is tempered, then*

$$\theta_\pi|_{\textit{nbhd of 0}} = \sum_i c_i \widehat{\mu_{\mathcal{O}_i}}|_{\textit{nbhd of 0}}.$$

Here $\mu_{\mathcal{O}_i} \in \mathcal{S}(\sqrt{-1}\mathfrak{g}_0^*)$, is the orbital integral along $\mathcal{O}_i$.

For an arbitrary generalized function $\Theta \in C^{-\infty}(M)$, we can define its wavefront set $\mathrm{WF}(\Theta) \subseteq \sqrt{-1}\mathrm{T}^*M$ as follows:

Definition 10.3. *For any $(x,\xi) \in \sqrt{-1}\mathrm{T}^*M$, $(x,\xi) \notin \mathrm{WF}(\Theta)$ if there exists a compactly supported smooth function φ on M such that $\varphi(x) \neq 0$, and an open conic neighborhood Γ of ξ such that for any $N \in \mathbb{N}$, there exists a constant $C_N > 0$ such that*

$$|\widehat{\varphi\Theta}(t\eta)| \leq C_N t^{-N}, \quad \forall t > 0, \eta \in \Gamma.$$

$\mathrm{WF}(\Theta)$ is a closed conic subset of $\sqrt{-1}\mathrm{T}^*M$ (singular direction of Θ).

In particular, we can define the wavefront set $\mathrm{WF}(\pi)$ of an irreducible Casselman-Wallach representation π of G as $\mathrm{WF}(\Theta_\pi) \cap (\sqrt{-1}\mathrm{T}_1^*(G))$, it is a closed conic G-invariant subset of $\sqrt{-1}\mathfrak{g}_0^*$. The GK-dimension of π is defined as

$$\mathrm{Dim}(\pi) = \frac{1}{2}\dim(\mathrm{WF}(\pi)).$$

θ_π is a real analytical function on $\mathfrak{g}_0^{\mathrm{rs}}$, and

$$\theta_\pi^*(X) := \lim_{t\to 0} t^{\mathrm{Dim}(\pi)}\theta_\pi(tX)$$

is still a real analytical function on $\mathfrak{g}_0^{\mathrm{rs}}$, where $\mathcal{B}$ is the flag variety of G. We can consider it as in $C^{-\infty}(\mathfrak{g}_0)$.

Theorem 10.4. *(Barbasch-Vogan) The asymptotic of θ_π defined above is a non-negative linear combination of Fourier transforms of nilpotent orbital integrals:*

$$\theta_\pi^* = \sum_i c_i \widehat{\mu_{\mathcal{O}_i}}.$$

Here $\mu_{\mathcal{O}_i} \in \mathcal{S}(\sqrt{-1}\mathfrak{g}_0^*)$, is the orbital integral along $\mathcal{O}_i$.

$\mathrm{WF}(\pi) = \bigcup_{c_i \neq 0} \overline{\mathcal{O}_i}$.

11. 2025.10.2

11.1. **Conormal bundle of singular subvariety.** Let $\mathcal{B}$ be the flag variety of G. $\overline{\mathcal{B}_y} \subsetneq \overline{\mathcal{B}_w}$ two schubert varieties, what is the relation between their conormal bundles $T^*_{\mathcal{B}_y}\mathcal{B}$ and $T^*_{\mathcal{B}_w}\mathcal{B}$?

Let X be a smooth algebraic variety over $\mathbb{C}$, $Z \subseteq X$ be a closed subvariety, define the conormal bundle $T^*_Z X$ of Z in X as follows:

$$T^*_Z X = \overline{T^*_{Z^{\mathrm{reg}}} X}.$$

For a singular point $z \in Z$ what is the fiber of $T^*_Z X$ at z?

Here is a simple example:

Example 11.1. *Let $X = \mathbb{C}^2$ and*

$$Z = \{xy = 0\} \subset X$$

be the union of the coordinate axes.

- *The regular locus is*

$$Z^{\mathrm{reg}} = (x\text{-}axis \setminus \{0\}) \cup (y\text{-}axis \setminus \{0\}).$$

- *The singular locus is*

$$S := Z \setminus Z^{\mathrm{reg}} = \{0\}.$$

Now, the fibre of the conormal bundle at the origin is

$$(T^*_Z X)_0 = \mathrm{Ann}(T_{x\text{-}axis}) \cup \mathrm{Ann}(T_{y\text{-}axis}) = \langle dy \rangle \cup \langle dx \rangle,$$

which is a union of two lines in $T^*_0\mathbb{C}^2$.

On the other hand,

$$(T^*_S X)_0 = (T^*_{\{0\}} X)_0 = T^*_0 X,$$

the entire cotangent plane.

Thus,

$$(T^*_Z X)_0 \subsetneq (T^*_{Z - Z^{\mathrm{reg}}} X)_0,$$

11.2. **Summary of paper** [KT84, The characteristic cycles of holonomic systems on a flag manifold]**.** Let G be a semisimple simply connected complex algebraic group. Define Steinberg variety:

$$Z = \{(x, B', B'') \in \mathcal{N} \times \mathcal{B} \times \mathcal{B} \,|\, x \in \mathrm{Lie}(B') \cap \mathrm{Lie}(B'')\} (= \{(x, B', B'') \in \mathcal{N} \times \mathcal{B} \times \mathcal{B} \,|\, \}).$$

$(= \{(x, B', B'') \in \mathcal{N} \times \mathcal{B} \times \mathcal{B} \,|\, x \in \mathfrak{n}'^{\perp} \cap \mathfrak{n}''^{\perp}\}$ more naturally.)

Here $\mathcal{N} \subseteq \mathfrak{g}$ is the nilpotent cone of $\mathfrak{g}$, $\mathcal{B}$ is the flag variety of G. It can be identify with the union of conormal bundles of all G-orbits in $\mathcal{B} \times \mathcal{B}$:

$$Z = \bigcup_{w \in W} T^*_{\mathcal{Y}_w}(\mathcal{B} \times \mathcal{B}),$$

which is a closed conic lagrangian subvariety of $T^*(\mathcal{B} \times \mathcal{B})$.

We also have the moment map:

$$\mu : Z \to \mathcal{N}, \quad (x, B', B'') \mapsto x.$$

By taking characteristic cycles, we get a map from the (complex) Grothendieck group of G-equivariant holonomic D-modules on $\mathcal{B} \times \mathcal{B}$ to the top Borel-Moore homology of Z:

$$\mathrm{CC} : K_0(\mathrm{Mod}^{\mathrm{Coh}}_{\Delta G}(\mathcal{D}_{\mathcal{B} \times \mathcal{B}})) \to H^{\mathrm{BM}}_{\mathrm{top}}(Z, \mathbb{C}).$$

We also have isomorphisms:

$$h : K_0(\mathrm{Mod}^{\mathrm{Coh}}_{\Delta G}(\mathcal{D}_{\mathcal{B}\times\mathcal{B}})) \to \mathbb{C}[W], \quad \mathcal{M}_w \mapsto w, \quad \mathcal{L}_w \mapsto a(w)$$

By Kazhdan-Lusztig, we have:

$$a(w) = \sum_{y\leq w} (-1)^{l(y)+l(w)} P_{y,w}(1) y.$$

There is a convolution product on $K_0(\mathrm{Mod}^{\mathrm{Coh}}_{\Delta G}(\mathcal{D}_{\mathcal{B}\times\mathcal{B}}))$ which makes h an isomorphism of algebras.

On

$$H^{\mathrm{BM}}_{\mathrm{top}}(Z, \mathbb{C}) \simeq \bigoplus_{w\in W} H^{\mathrm{BM}}_{\mathrm{top}}(\overline{T^*_{\mathcal{Y}_w}(\mathcal{B}\times\mathcal{B})}, \mathbb{C}) \simeq \bigoplus_{w\in W} \mathbb{C}[\overline{T^*_{\mathcal{Y}_w}(\mathcal{B}\times\mathcal{B})}],$$

In the paper of Kazhdan-Lusztig [KL80], they use topological method to construct a $W\times W$ action on $H^{\mathrm{BM}}_{\mathrm{top}}(Z, \mathbb{C})$.

Moreover there is a $W\times W$ module decomposition:

$$H^{\mathrm{BM}}_{\mathrm{top}}(Z, \mathbb{C}) \simeq \bigoplus_{\mathcal{O}} H^{\mathrm{BM}}_{\mathrm{top}}(Z_{\mathcal{O}}, \mathbb{C}),$$

where $\mathcal{O}$ runs over all nilpotent orbits in $\mathfrak{g}^*$, $Z_{\mathcal{O}} = \mu^{-1}(\mathcal{O})$. There is also an isomorphism of $W\times W$-modules:

$$H^{\mathrm{BM}}_{\mathrm{top}}(Z_{\mathcal{O}}, \mathbb{C}) \simeq (H^{\mathrm{BM}}_{\mathrm{top}}(\mathcal{B}_x, \mathbb{C}) \otimes H^{\mathrm{BM}}_{\mathrm{top}}(\mathcal{B}_x, \mathbb{C}))_{A_G(x)},$$

($A_G(x)$ coinvariants) where $x\in\mathcal{O}$, $\mathcal{B}_x = \{B'\in\mathcal{B} \,|\, x\in \mathrm{Lie}(B')\}$ (the Springer fiber), $A_G(x) = \mathrm{Stab}_G(x)/\mathrm{Stab}_G(x)^0$.

By classical Springer theory, $H^{\mathrm{BM}}_{\mathrm{top}}(\mathcal{B}_x, \mathbb{C})$ is a representation of $W\times A_G(x)$, and we have the following decomposition:

$$H^{\mathrm{BM}}_{\mathrm{top}}(\mathcal{B}_x, \mathbb{C}) \simeq \bigoplus_{\rho\in\widehat{A_G(x)}} \tau(x,\rho)\otimes\rho.$$

Here $\tau(x,\rho)$ is an irreducible representation of W or zero.

So we have a decomposition of $W\times W$-modules:

$$H^{\mathrm{BM}}_{\mathrm{top}}(Z_{\mathcal{O}}, \mathbb{C}) \simeq \bigoplus_{\rho\in\widehat{A_G(x)}} \tau(x,\rho)\otimes\tau(x,\rho).$$

Combine the above together, we finally have the following decomposition:

$$H^{\mathrm{BM}}_{\mathrm{top}}(Z_{\mathcal{O}}, \mathbb{C}) \simeq \bigoplus_{\rho\in\widehat{A_G(x)}} \tau(x,\rho)\otimes\tau(x,\rho), \quad H^{\mathrm{BM}}_{\mathrm{top}}(Z, \mathbb{C}) \simeq \bigoplus_{\mathcal{O}} H^{\mathrm{BM}}_{\mathrm{top}}(Z_{\mathcal{O}}, \mathbb{C}).$$

For any $w\in W$, there is a unique nilpotent orbit $\mathrm{St}(w)$ such that $\mu(\overline{T^*_{\mathcal{Y}_w}(\mathcal{B}\times\mathcal{B})}) = \overline{\mathrm{St}(w)}$. The map $w\mapsto \mathrm{St}(w)$ defines a surjective(?) map from W to the set of nilpotent orbits in $\mathfrak{g}^*$ (this map can be computed via Robinson-Schensted algorithm).

Corollary 11.2. *For any nilpotent orbit $\mathcal{O}$, there is an isomorphism of $W\times W$-modules:*

$$\bigoplus_{w\in W, \mathrm{St}(w)=\mathcal{O}} \mathbb{C}[\overline{T^*_{\mathcal{Y}_w}(\mathcal{B}\times\mathcal{B})}] \simeq \bigoplus_{\rho\in\widehat{A_G(\mathcal{O})}} \tau(\mathcal{O},\rho)\otimes\tau(\mathcal{O},\rho).$$

Theorem 11.3. *(Main theorem of* [KT84]*) The map*

$$\mathrm{CC} : K_0(\mathrm{Mod}^{\mathrm{Coh}}_{\Delta G}(\mathcal{D}_{\mathcal{B}\times\mathcal{B}})) \to H^{\mathrm{BM}}_{\mathrm{top}}(Z, \mathbb{C}),$$

is an isomorphism and $W \times W$-equivariant.

Define $b(w) = h \circ \mathrm{CC}^{-1}([\overline{T^*_{\mathcal{Y}_w}\mathcal{B}\times\mathcal{B}}])$. Now we have three basis of $\mathbb{C}[W]$:

$$\{w\}_{w\in W}, \quad \{a(w)\}_{w\in W}, \quad \{b(w)\}_{w\in W}.$$

It was conjectured by Kashiwara and Tanisaki that: $a(w) = b(w)$ when $G = \mathrm{SL}_n(\mathbb{C})$ which means the characteristic cycle of $\mathcal{L}_w$ is irreducible, but unfortunately it is not true. Conterexample exists when $G = \mathrm{SL}_8(\mathbb{C})$, where Kashwara-Saito singularities appears in some Schubert varieties.

12. 2025.10.5

The Springer representation

Let G be a connected complex reductive group, K a symmetric subgroup of G. We can define the Springer representation of W on $H_c^\bullet(T_K^*\mathcal{B}, \mathbb{Q})$.

Recall that we have the following diagram:

$$\begin{tikzcd} \widetilde{\mathcal{N}} \arrow[r, hook] \arrow[d, two heads, "p_{\mathcal{N}}"] & \widetilde{\mathfrak{g}} \arrow[d, two heads, "p"] & \widetilde{\mathfrak{g}^{\mathrm{rs}}} \arrow[l, hook'] \arrow[d, two heads, "p_{\mathrm{rs}}"] \\ \mathcal{N} \arrow[r, hook] & \mathfrak{g} & \mathfrak{g}^{\mathrm{rs}} \arrow[l, hook'] \end{tikzcd}$$

In this diagram p_{rs} is a W-torsor. So we have a natural W-action on $p_{\mathrm{rs},*}\underline{\mathbb{Q}}_{\widetilde{\mathfrak{g}^{\mathrm{rs}}}}$ and hence on $\mathrm{IC}(\mathfrak{g}^{\mathrm{rs}}, \mathcal{L})$, where $\mathcal{L} = (p_{\mathrm{rs}})_*\underline{\mathbb{Q}}_{\widetilde{\mathfrak{g}^{\mathrm{rs}}}}$. There is a canonical isomorphism of perverse sheaves $\mathrm{IC}(\mathfrak{g}^{\mathrm{rs}}, \mathcal{L}) \simeq \mathbb{R}p_*\underline{\mathbb{Q}}_{\widetilde{\mathfrak{g}}}[\dim(\mathfrak{g})]$, so there is a natural W-action on $\mathbb{R}p_*\underline{\mathbb{Q}}_{\widetilde{\mathfrak{g}}}$. By proper base change, we have

$$\mathbb{R}p_{\mathcal{N},*}\underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}} \simeq \mathbb{R}p_*\underline{\mathbb{Q}}_{\widetilde{\mathfrak{g}}})|_{\mathcal{N}}. \quad (\mathrm{Spr} = \mathbb{R}p_{\mathcal{N},*}\underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}}[\dim \mathcal{N}] \textit{ called The Springer sheaf})$$

So there is a natural W-action on $\mathbb{R}p_{\mathcal{N},*}\underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}}$ (actually $\mathrm{End}_{\mathrm{Perv}(\mathcal{N})}(\mathrm{Spr}) = \mathbb{Q}[W]$) and hence on $H_c^\bullet(T^*\mathcal{B}, \mathbb{Q}) \simeq \mathbb{R}\Gamma_c(\mathcal{N}, \mathbb{R}p_{\mathcal{N},*}\underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}})$ and $H_c^\bullet(\mathcal{B}_x, \mathbb{Q}) \simeq \mathbb{R}\Gamma_c(\{x\}, \mathbb{R}p_{\mathcal{N},*}\underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}})$. (action on Springer fiber)

Since $T_K^*\mathcal{B} = p_{\mathcal{N}}^{-1}(\mathfrak{p})$, we also have a natural W-action on

$$H_c^\bullet(T_K^*\mathcal{B}, \mathbb{Q}) \simeq \mathbb{R}\Gamma_c(\mathfrak{p} \cap \mathcal{N}, (\mathbb{R}p_{\mathcal{N},*}\underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}})|_{\mathfrak{p}\cap\mathcal{N}}).$$

Finally, there is a W-action on $(H_c^{2d}(T_K^*\mathcal{B}, \mathbb{Q}))^* = \bigoplus_{\mathcal{O}\in K\backslash\mathcal{B}} \mathbb{Q}[\overline{T_{\mathcal{O}}^*\mathcal{B}}]$.

Theorem 12.1 (Tanisaki)**.** *The map:*

$$\mathrm{CC} : K_0(\mathrm{Mod}_K^{\mathrm{Coh}}(\mathcal{D}_{\mathcal{B}})) \to (H_c^{2d}(T_K^*\mathcal{B}, \mathbb{Q}))^* = H_{2d}^{\mathrm{BM}}(T_K^*\mathcal{B}, \mathbb{Q}),$$

is surjective and W-equivariant.

13. 2025.10.9

$X = \mathrm{Spec}(R)$ smooth affine with coordinate $x_1, \ldots, x_n$, M a finitely generated $\mathcal{D}_X$-module.

Definition 13.1. *M is called holonomic if* $\dim(\mathrm{Ch}(M)) = n$.

Corollary 13.2. *If M is holonomic, then $\mathrm{Ch}(M)$ is lagrangian in T^*X, and* $\mathrm{CC}(M) = \sum_{E_i \subseteq X irr} n_i [\overline{T^*_{E_i} X}]$.

The de-Rham complex and analytification:

$$\mathcal{K}(M; \partial_1) = [\, M \xrightarrow{\partial_1} M \,],$$

$$\mathcal{K}(M; \partial_1, \partial_2) = [\, M \xrightarrow{(\partial_1, \partial_2)} M \oplus M \xrightarrow{(\partial_2, -\partial_1)} M \,].$$

We can define $\mathcal{K}(M; \partial_1, \ldots, \partial_n)$ inductively.

There is another way to define the de-Rham complex, independent of the choice of coordinates:

$$\mathrm{DR}(M) = [M \xrightarrow{\nabla} \Omega^1_X \otimes M \xrightarrow{\nabla} \cdots \xrightarrow{\nabla} \Omega^n_X \otimes M].$$

Example 13.3. *Let $X = \mathbb{C}$, $M = \mathbb{C}[x, x^{-1}] \cdot x^\alpha$, $\alpha \in \mathbb{C}$, $\partial_x x^\alpha = \alpha \frac{1}{x} x^\alpha$. Then $M_\alpha = \mathcal{D}_X / \mathcal{D}_X (x\partial_x - (\alpha - k))$. To get the* multiplicities *(the characteristic cycle), we examine the primary decomposition of the defining principal-symbol ideal:*

$$(x\xi) = (x) \cap (\xi).$$

Each factor appears with exponent 1, and the associated graded module localized at a generic point of either component has length 1. Concretely:

- *Localize at the generic point of $V(\xi)$ (i.e. at the prime (ξ)). In that localization, ξ is nonunitally small and x is a unit, so*

$$(\mathrm{gr}\, M_\alpha)_{(\xi)} \cong \mathbb{C}[x, \xi]_{(\xi)} / (\xi)$$

  which is a copy of $\mathbb{C}[x]_{(0)}$ of length 1 over the local ring — so the multiplicity is 1 on the zero section.
- *Localize at the generic point of $V(x)$ (i.e. at (x)). There ξ is a unit and*

$$(\mathrm{gr}\, M_\alpha)_{(x)} \cong \mathbb{C}[x, \xi]_{(x)} / (x)$$

  which is a copy of $\mathbb{C}[\xi]_{\mathrm{gen}}$ of length 1 — so the multiplicity is 1 on the conormal $T^*_0 X$.

Hence both components appear with multiplicity 1.

Therefore the characteristic cycle is

$$\boxed{\mathrm{CC}(M_\alpha) \;=\; [T^*_X X] \;+\; [T^*_0 X].}$$

This calculation is independent of α, because the parameter α contributes only lower-order terms that do not affect the principal symbol.

Now consider the de-Rham complex of M_α:

$$\mathrm{DR}(M_\alpha) = [M_\alpha \xrightarrow{\partial_x} M_\alpha].$$

It is exact, this is bad. To fix this, we need to analytify M_α:

$$M_\alpha^{\mathrm{an}} = \mathcal{O}_{X^{\mathrm{an}}} \otimes_{\mathbb{C}[x]} M_\alpha = \mathcal{O}_{X^{\mathrm{an}}}[x^{-1}] \cdot x^\alpha.$$

Example 13.4. *$N = \mathbb{C}[\delta]$ where δ is the delta function at 0, $\partial_x \delta = 0$. Then $N = \mathcal{D}_X / \mathcal{D}_X x$. The characteristic cycle is $\mathrm{CC}(N) = [T_0^* X]$. The de-Rham complex $\mathrm{DR}(N) = [N \xrightarrow{\partial_x} N]$ is not exact, and $H^0(\mathrm{DR}(N)) = \mathbb{C}\delta$. The analytification $N^{\mathrm{an}} = \mathbb{C}\{x\} \cdot \delta$. The de-Rham complex $\mathrm{DR}(N^{\mathrm{an}}) = [N^{\mathrm{an}} \xrightarrow{\partial_x} N^{\mathrm{an}}]$ is still not exact, and $H^0(\mathrm{DR}(N^{\mathrm{an}})) = \mathbb{C}\delta$.*

Remark 13.5. *The two examples above consists of all regular singularities in some sence.*

14. 2025.10.13

Lemma 14.1. *Let G be a connected Lie group, $\widetilde{G}$ is a double cover. Then the following are equivalent:*

- *$\widetilde{G}$ split;*
- *$\widetilde{G}$ is disconnected;*
- *there exists a genuine representation of $\widetilde{G}$ which is trivial on the identity component $\widetilde{G}_0$.*

Proof. Only need to notice that $\widetilde{G}_0 \to G$ is a covering map of degree 1 or 2. $\square$

There is a same statement for p-adic groups in [**?**].

Lemma 14.2. *Let G be an arbitrary Lie group, $\pi : \widetilde{G} \to G$ a double cover. Then the following are equivalent:*

- *there exists a genuine finite dimensional representation of $\widetilde{G}$ which is trivial on the identity component $\widetilde{G}_0$;*
- *$z \notin \widetilde{G}_0$;*
- *$z \notin (\pi^{-1}(G_0))_0$;*
- *there exist a genuine finite dimensional representation of $\pi^{-1}(G_0)$ which is trivial on the identity component $(\pi^{-1}(G_0))_0$;*
- *$\pi^{-1}(G_0)$ is disconnected;*
- *$\pi^{-1}(G_0)$ is split.*

Here z is the nontrivial element in $\ker(\pi)$.

If in addition the double cover $\widetilde{G}$ is induced by a square root of a character $\gamma : G \to \mathbb{C}^\times$, i.e. $\widetilde{G} = \{(g, \epsilon) \in G \times \mathbb{C}^\times \mid \epsilon^2 = \gamma(g)\}$, these are also equivalent to:

- *γ has a square root.*

Proof. Only need to notice that $(\pi^{-1}(G_0))_0 = \widetilde{G}_0$, together with results in [**?**]. $\square$

15. 2025.10.14

$X = \mathrm{Spec}(R)$ smooth affine with coordinate $x_1, \ldots, x_n$, M is an integrable connection.

Theorem 15.1. *$M^{an} = L \otimes_{\mathbb{C}} \mathcal{O}_{X^{an}}$ for some local system L on X^{an}.*

Proof. We know that M is locally free, suppose $M|_U$ is freely generated by $m_1, \cdots, m_n$, suppose $\partial_n \underline{m} = \chi \underline{m}$ where χ is a matrix of holomorphic functions on U. Locally we can expand χ with respect to x_n:

$$\chi = \sum_{i=0}^{\infty} \chi_i(x_1, \ldots, x_{n-1}) x_n^i.$$

Take $\Psi = \sum_{i=1}^{\infty} \frac{1}{i+1} \chi_i(x_1, \ldots, x_{n-1}) x_n^i$, so $\partial_n \Psi = \chi$. Take $\Phi = \exp(-\Psi)$ which is invertible, and set $\underline{s} = \Phi \underline{m}$.

So

$$\partial_n s = \partial_n(\Phi) \cdot \underline{m} + \Phi \cdot (\partial_n \underline{m}) = 0.$$

There is a small open subset $V \subset U$ such that $M^{an}|_V = \bigoplus_{i=1}^r \mathcal{O}_V s_i$, take

$$\begin{aligned} M_1 &= \ker(M^{an}|_V \xrightarrow{\partial_n} M^{an}|_V) \\ &= \{\sum_{i=1}^r f_i s_i \in M^{an}|_V \mid \partial_n(\sum_{i=1}^r f_i s_i) = 0, \forall i\} \\ &= \{\sum_{i=1}^r f_i s_i \in M^{an}|_V \mid \partial_n f_i = 0, \forall i\} \\ &= \bigoplus_{i=1}^r \mathcal{O}_{V_1} s_i, \quad V_1 = V \cap \{x_n = 0\}. \end{aligned}$$

Continue this process, we can finally get a subsheaf

$$M_n|_V = \bigoplus_{i=1}^n \mathbb{C} s_i.$$

of M^{an}, where s_i are horizontal sections. So $M^{an} = L \otimes_{\mathbb{C}} \mathcal{O}_{X^{an}}$ where L is the local system generated by s_i.

□

Remark 15.2. *The proof actually implies that integrable connections are locally generated by horizontal sections, and the sheaf of horizontal sections is a local system.*

Corollary 15.3. $\mathrm{DR}(M) = L$

Proof. $H^0(\mathrm{DR}(M)) \simeq \ker(M \xrightarrow{\nabla} \Omega_X^1 \otimes M) = L$. All the other cohomologies vanish because of Poincaré lemma, since this complex is locally de-Rham complex. □

Remark 15.4. *This means that de Rham complex of M gives a locally free resolution of L. So the hypercohomology of $\mathrm{DR}(M)$ computes the cohomology of L.*

16. 2025.10.16

Kashiwara's equivalence

Y smooth affine $/\mathbb{C}$ with coordinate $x_1, \cdots, x_n, y_1, \cdots, y_l$, $X = \{y_1 = \cdots = y_l = 0\} \subset Y$, X is also smooth. $I = (y_1, \ldots, y_l)$, $\mathcal{D}_Y$ the ring of differential operators on Y, $\mathcal{D}_X$ the ring of differential operators on X. $\mathrm{Mod}_{qc}(\mathcal{D}_X)$: the category of quasi-coherent $\mathcal{D}_X$ modules, $\mathrm{Mod}_{qc}^X(\mathcal{D}_Y)$: the category of quasi-coherent $\mathcal{D}_Y$ modules supported on X.

Theorem 16.1. *(Kashiwara's equivalence) The functor:*

$$i_+ : \mathrm{Mod}_{qc}(\mathcal{D}_X) \to \mathrm{Mod}_{qc}^X(\mathcal{D}_Y), \quad M \mapsto \mathcal{D}_{Y \leftarrow X} \otimes_{\mathcal{D}_X} M = M[\partial_{y_1}, \cdots, \partial_{y_l}],$$

is an equivalence of categories. Its quasi-inverse is given by:

$$i^\sharp : \mathrm{Mod}_{qc}^X(\mathcal{D}_Y) \to \mathrm{Mod}_{qc}(\mathcal{D}_X), \quad N \mapsto \ker(N \longrightarrow^y N).$$

Proof. There is a proof in [HTT08], I don't want to repeat it here. □

Remark 16.2. *In the case of quasi-coherent sheaves, the equivalence does not hold. For example, Let $Y = \mathrm{Spec}(R)$, $X = \mathrm{Spec}(R/I) \subseteq Y$ a closed subscheme defined by ideal I. Then the functor:*

$$i^* : \mathrm{Mod}_{qc}^X(\mathcal{O}_Y) \to \mathrm{Mod}_{qc}(\mathcal{O}_X), \quad N \mapsto R/I \otimes_R N.$$

is not an equivalence of categories. Since if M is an R-module, then M/I^2M is supported on X, but it does not defined over R/I.

Corollary 16.3. *Let M be a quasi-coherent $\mathcal{D}_X$-module, if $Ch(M) = T_Y^*X$ for an irreducible closed subvariety $X \subset Y$, then $M = i_+(N)$ for some integrable connection N on Y.*

$$\begin{tikzcd} \mathrm{Mod}_{qc}(\mathcal{D}_X) \arrow[r, shift left, "i_+"] & \mathrm{Mod}_{qc}^X(\mathcal{D}_Y) \arrow[l, shift left, "i^\sharp"] \end{tikzcd}$$

$$\begin{tikzcd} \boxed{\textbf{Orbital integrals of } G} \arrow[loop above] \arrow[r, dashed, leftrightarrow, "f"'] \arrow[d, dashed, leftrightarrow, "\text{Transfer}"] & \boxed{\textbf{L-packets of } G} \arrow[loop above] \arrow[d, dashed, leftrightarrow, "\text{Endoscopic}"'] \\ \boxed{\textbf{Orbital integrals of } H} \arrow[loop below] \arrow[r, dashed, leftrightarrow, "f_H"'] & \boxed{\textbf{L-packets of } H} \arrow[loop below] \end{tikzcd}$$

17. 2025.10.19

Degenerate Whittaker model

G be a finite extension of a reductive algebraic group defined over local field F, $\mathfrak{g}$ be its Lie algebra, fix a nontrivial character $\chi : F \to \mathbb{C}^1$.

A Whittaker pair is a pair (S, φ) where $S \in \mathfrak{g}$ is a rational semi-simple element, $\varphi \in \mathfrak{g}^*$ such that $\mathrm{ad}^*(S)(\varphi) = -2\varphi$. Given such a Whittaker pair, we can define a degenerate Whittaker model $\mathcal{W}_{S,\varphi}$ as follows: Let $\mathfrak{g} = \bigoplus_{r\in\mathbb{Q}} \mathfrak{g}_r^S$ be the eigenspace decomposition with respect to $\mathrm{ad}(S)$, define

$$\mathfrak{u} = \mathfrak{g}_{\geq 1} = \bigoplus_{r\geq 1} \mathfrak{g}_r^S.$$

Define an anti-symmetric form ω_φ on $\mathfrak{u}$ by

$$\omega_\varphi(X, Y) = \varphi([X, Y]), \quad X, Y \in \mathfrak{u}.$$

Let $\mathfrak{n}$ be the kernel of ω_φ, and $\mathfrak{n}' = \mathfrak{n} \cap \ker(\varphi)$. Then $\mathfrak{u}/\mathfrak{n}$ is a symplectic space with symplectic form induced by ω_φ, and $\mathfrak{u}/\mathfrak{n}'$ is a Heisenberg Lie algebra with center $\mathfrak{n}/\mathfrak{n}'$. Corresponding groups are defined by exponentiation:

$$U = \exp(\mathfrak{u}), \quad N = \exp(\mathfrak{n}), \quad N' = \exp(\mathfrak{n}').$$

If $\varphi \neq 0$, there is a unique irreducible representation $(\rho_\varphi, \mathcal{S}_\varphi)$ of U/N' with central character $\chi_\varphi : N/N' \to \mathbb{C}^\times$ defined by

$$\chi_\varphi(\exp(X)) = \chi(\varphi(X)), \quad X \in \mathfrak{n}.$$

Define the degenerate Whittaker model:

$$\mathcal{W}_{S,\varphi} = \mathrm{Ind}_U^G(\rho_\varphi).$$

If $\varphi = 0$, define $\mathcal{W}_{S,0} = \mathrm{Ind}_U^G(1_U)$.

Remark 17.1. *In the special case when (S, φ) comes from an $\mathfrak{sl}_2$-triple (e, h, f) with $S = h$ and φ corresponding to f via the Killing form, we can see that $\mathfrak{n} = \mathfrak{g}_{\geq 2}$. Then $\mathcal{W}_f = \mathcal{W}_{S,\varphi}$ is called a generalized Whittaker model associated to the nilpotent orbit of f.*

Affine transformation group of real line

$P_2(\mathbb{R}) = \mathbb{R}^\times \ltimes \mathbb{R}$, with multiplication:

$$(a, b) \cdot (a', b') = (aa', b + ab').$$

Its Lie algebra is $\mathfrak{p}_2(\mathbb{R}) = \mathbb{R} \times \mathbb{R}$ with bracket:

$$[(x_1, y_1), (x_2, y_2)] = (0, x_1y_2 - x_2y_1).$$

18. 2025.10.21

Spectral Sequences of Filtrated Complexes

$(F^\bullet, d)$ be a filtered complex, i.e. a complex $F^\bullet$ with an increasing filtration of subcomplexes:

$$\cdots \subseteq F_j^\bullet \subseteq F_{j+1}^\bullet \subseteq \cdots \subseteq F^\bullet.$$

Such that $d(F_j^i) \subseteq F_j^{i+1}$, and the filtration is exhaustive and separated:

$$\bigcup_{j\in\mathbb{Z}} F_j^i = F^i, \quad \bigcap_{j\in\mathbb{Z}} F_j^i = 0.$$

For any triple (i,j,k), define:

- $Z_j^i(k) := \{u \in F_j^i \mid du \in F_{j-k}^{i+1}\}$;
- $B_j^i(k) := d(F_{j+k-1}^{i-1}) \cap F_j^i$;
- $E_j^i(k) := \frac{Z_j^i(k)+F_{j-1}^i}{B_j^i(k)+F_{j-1}^i}$.

So $d(Z_j^i(k)) \subseteq Z_{j-k}^{i+1}(k)$, which induces a differential $d_k : E_j^i(k) \to E_{j-k}^{i+1}(k)$.

We also define $E_k^{p,q} := E_{-p}^{p+q}(k)$.

Example 18.1. *$E_j^i(0) = \frac{F_j^i}{F_{j-1}^i}$, so $E_0^{p,q} = \mathrm{Gr}_{-p}F^{p+q}$.*

$E_j^i(1) = H^i(\mathrm{Gr}_j F^\bullet)$, so $E_1^{p,q} = H^{p+q}(\mathrm{Gr}_{-p}F^\bullet)$. The first page of the spectral sequence is just the cohomology of the associated graded complex.

Lemma 18.2. *$H^i(E_j^\bullet(k)) = E_j^i(k+1)$.*

Definition 18.3. *The spectral sequence associated to the filtered complex $(F^\bullet, d)$ is defined to be the collection of complexes $\{(E_\bullet^\bullet(k), d_k)\}_{k\geq 0}$.*

They convergent to the cohomology of the original complex: $E_\infty^{p,q} \simeq \mathrm{Gr}_{-p}H^{p+q}(F^\bullet)$, if for any i,j, there exist $k \gg 0$ such that $E_j^i(k) = \mathrm{Gr}_j H^i(F^\bullet)$.

Definition 18.4. *If there exist a $w \in \mathbb{Z}_{\geq 0}$ such that for any i,j, $F_j^i \cap d(F^{i-1}) \subseteq d(F_{j+w}^{i-1})$. Then the filtration is called w-regular.*

Theorem 18.5. *If $F^\bullet$ is w-regular, then the spectral sequence associated to $(F^\bullet, d)$ converges to the cohomology of the original complex at page $w+1$, i.e. for any i,j, $E_j^i(w+1) \simeq \mathrm{Gr}_j H^i(F^\bullet)$.*

Example 18.6. *$K^\bullet$ is a chain complex, F a left exact functor, then there is a natural spectral sequence:*

$$E_2^{p,q} = \mathbb{R}^{p+q}F(K^q) \Rightarrow \mathbb{R}^{p+q}F(K^\bullet).$$

(maybe:

$$E_1^{p,q} = \mathbb{R}^{p}F(K^q) \Rightarrow \mathbb{R}^{p+q}F(K^\bullet)?$$

)

Because we can use the Cartan-Eilenberg resolution to resolve $K^\bullet$, which is given by:

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & K^0 \arrow[r] \arrow[d] & I^{0,0} \arrow[r] \arrow[d] & I^{0,1} \arrow[r] \arrow[d] & \cdots \\
0 \arrow[r] & K^1 \arrow[r] \arrow[d] & I^{1,0} \arrow[r] \arrow[d] & I^{1,1} \arrow[r] \arrow[d] & \cdots \\
0 \arrow[r] & K^2 \arrow[r] \arrow[d] & I^{2,0} \arrow[r] \arrow[d] & I^{2,1} \arrow[r] \arrow[d] & \cdots \\
 & \vdots & \vdots & \vdots &
\end{tikzcd}$$

Then we can form a total complex $\mathrm{Tot}(I^{\bullet,\bullet})$ *with a natural filtration, and apply the functor* F *to get a filtered complex* $(F(\mathrm{Tot}(I^{\bullet,\bullet})), d)$. *The spectral sequence associated to this filtered complex is just the spectral sequence above.*

A special case is the Hodge spectral sequence, when X *is a smooth algebraic variety over* $\mathbb{C}$, $K^\bullet = \Omega_X^\bullet$ *the de-Rham complex, and* $F = \mathbb{R}\Gamma(X, -)$ *the derived global section functor, the spectral sequence is:*

$$E_1^{p,q} = H^q(X, \Omega_X^p) \Rightarrow H_{\mathrm{dR}}^{p+q}(X).$$

19. 2025.10.23

Lemma 19.1. *Let H be a Lie group acting freely and properly on a smooth manifold M, then for any smooth manifold T, there is a natural isomorphism:*

$$\mathrm{Hom}_{sm}(T, M/H) \simeq \frac{\{(P \to^{\pi} T, f : P \to M) | \pi \text{ a principle } H\text{-bundle, } f \text{ is } H\text{-equivariant}\}}{\text{equivalence}}.$$

Proof. Given a smooth map $g : T \to M/H$, we can form the following Cartesian diagram:

$$\begin{tikzcd} P \arrow[r, "f"] \arrow[d, "\pi"'] & M \arrow[d] \\ T \arrow[r, "g"'] & M/H \end{tikzcd}$$

Since H acts freely and properly on M, $\pi : P \to T$ is a principle H-bundle, and $f : P \to M$ is H-equivariant.

Conversely, given a principle H-bundle $\pi : P \to T$ and an H-equivariant map $f : P \to M$, we can define a smooth map $g : T \to M/H$ by sending $t \in T$ to the orbit of $f(p)$ where $p \in \pi^{-1}(t)$. □

Remark 19.2. *Algebraic geometers use this to define the quotient stack $[M/H]$ even if the action is not free or proper, but $\mathrm{Hom}(T, [M/H])$ need to be a groupoid, not just a set of equivalence classes.*

20. 2025.10.29

Unipotent Arthur packets and special unipotent representation

Let G be a connected reductive algebraic group over $\mathbb{R}$, fix a weak E-group $\check{G}^\Gamma$. Which is an extension:

$$1 \to \check{G} \to \check{G}^\Gamma \to \mathrm{Gal}(\mathbb{C}/\mathbb{R}) \to 1.$$

Here $\check{G}$ is the complex dual group of G. An L-parameter is a continuous homomorphism:

$$\phi : W_\mathbb{R} \to \check{G}^\Gamma,$$

such that the composition $W_\mathbb{R} \to {}^L\check{G} \to \mathrm{Gal}(\mathbb{C}/\mathbb{R})$ is the natural projection and $\phi(\mathbb{C}^*)$ consists of semisimple elements of $\check{G}$.

Given an L-parameter ϕ, we can attach the following data:

- two semisimple elements $\lambda, \mu \in \check{G}$, defined by $\phi(e^t) = \exp(\lambda t + \mu \bar{t})$;
- $y = \exp(\pi i \lambda)\phi(j) \in \check{G}^\Gamma - \check{G}$.

The L-parameter ϕ is called *tempered* if the closure of $\phi(W_\mathbb{R})$ in the complex analytic topology is compact. This is equivalent to say that $\lambda + \mathrm{Ad}(y)\lambda$ is an elliptic element of $\check{\mathfrak{g}}$.

Fix a geometric parameter (y, Λ), it lies in a variety $X_y(\mathcal{O}, \check{G}^\Gamma) \simeq \check{G} \times^{K(y)} (\check{G}(\Lambda)_0/P(\Lambda))$. The $\check{G}$-equivariant microlocal geometry on $X_y(\mathcal{O}, \check{G}^\Gamma)$ is in some sense equivalent to the $K(y)$-equivariant microlocal geometry on $\check{G}(\lambda)_0/P(\Lambda)$.

Roughly speaking, there is a bijection between:

$$\boxed{\text{representation with minimal GK-dim}}$$
$$\updownarrow$$
$$\boxed{\text{parameters supported on regular orbits in } \check{G}(\Lambda)_0/P(\Lambda)}$$

Definition 20.1. *An Arthur parameter is a homomorphism:*

$$\psi : W_\mathbb{R} \times SL(2, \mathbb{C}) \to \check{G}^\Gamma,$$

such that the restriction to $W_\mathbb{R}$ (ψ_0) is a tempered L-parameter, and the restriction to $SL(2, \mathbb{C})$ (ψ_1) is algebraic.

For an arthur parameter ψ, we can define an L-parameter ϕ_ψ by:

$$\phi_\psi(w) = \psi(w, \begin{pmatrix} |w|^{1/2} & 0 \\ 0 & |w|^{-1/2} \end{pmatrix}).$$

Let $y_1 = \psi_1 \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \in \check{G}$, and $\lambda_1 = d\psi_1 \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} \in \check{\mathfrak{g}}$, (y_0, λ_0) is the parameter defined by the restriction of ψ to $W_\mathbb{R}$.

Then we have $y = y_0 y_1, \lambda = \lambda_0 + \lambda_1$, since $\phi_\psi(e^t) = \exp(\lambda_0 t + \mu_0 \bar{t})\psi_1 \begin{pmatrix} |e^t| & 0 \\ 0 & |e^{-t}| \end{pmatrix}$, and

$$\psi_1 \begin{pmatrix} |e^t| & 0 \\ 0 & |e^{-t}| \end{pmatrix} = \exp\left(t \cdot d\psi_1 \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} + \bar{t} \cdot d\psi_1 \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}\right) = \exp(\lambda_1 t + \lambda_1 \bar{t}).$$

So $\phi_\psi(e^t) = \exp((\lambda_0 + \lambda_1)t + (\mu_0 + \lambda_1)\bar{t})$, hence $\lambda = \lambda_0 + \lambda_1$.

And

$$y = \exp(\pi i\lambda)\phi_\psi(j) = \exp(\pi i\lambda_0)\exp(\pi i d\psi_1\begin{pmatrix}\frac{1}{2} & 0\\ 0 & -\frac{1}{2}\end{pmatrix})\psi_0(j) = \exp(\pi i\lambda_0)\psi_1\begin{pmatrix}i & 0\\ 0 & -i\end{pmatrix}\psi_0(j) = y_0y_1.$$

In addition of the two semisimple elements λ, μ and y defined as above, we can also define a nilpotent element $E_\psi \in \check{\mathfrak{g}}$ by:

$$E_\psi = d\psi_1\begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix} \in \check{\mathfrak{g}}.$$

E_ψ enjoys the following properties:

- $\mathrm{Ad}(y)E_\psi = -E_\psi$, this means $E_\psi \in \mathfrak{s}(y)$;
- $[\lambda, E_\psi] = E_\psi$, in other words, $E_\psi \in \check{\mathfrak{g}}(\Lambda)_1 \subseteq \mathfrak{n}(\lambda)$ where $\check{\mathfrak{g}}(\Lambda) = \bigoplus_{r\in\mathbb{Z}}\check{\mathfrak{g}}_r$ is the eigenspace decomposition with respect to $\mathrm{ad}(\lambda)$.

$$\begin{array}{c}\text{Langlands parameter } (y,\lambda)\\ \rightsquigarrow\\ S = \check{G}\times^{K(y)} eP(\Lambda) \subseteq X_y(\mathcal{O}, \check{G}^\Gamma) = \check{G}\times^{K(y)}\check{G}(\Lambda)/P(\Lambda)\\ \rightsquigarrow\\ S_K = K(y)\cdot(eP(\Lambda)) \subseteq \check{G}(\Lambda)_0/P(\Lambda)\end{array}$$

Cotangent space at the base point:

$$T^*_{eP(\Lambda)}(\check{G}(\Lambda)_0/P(\Lambda)) \simeq \mathfrak{n}(\lambda).$$

Conormal space at the base point of the $K(y)$-orbit:

$$T^*_{K(y),eP(\Lambda)}(\check{G}(\Lambda)_0/P(\Lambda)) \simeq \mathfrak{s}(y)\cap\mathfrak{n}(\lambda).$$

So we can view E_ψ as an element in the conormal space $T^*_{K(y),eP(\Lambda)}(\check{G}(\Lambda)_0/P(\Lambda))$ also in $T^*_{\check{G},\phi_\psi}(X(\check{G}^\Gamma))$.

Lemma 20.2. *Consider the partial Springer resolution*

$$\mu : T^*(\check{G}(\Lambda)_0/P(\Lambda)) \simeq \check{G}(\Lambda)_0\times^{P(\Lambda)}\mathfrak{n}(\lambda) \to \mathcal{N}_{\mathcal{P}(\Lambda)},$$

$E_\psi \in \mathcal{N}_{\mathcal{P}(\Lambda)}\cap\mathfrak{s}(y)$ is a regular element (belong to the Richardson orbit), and the fiber $\mu^{-1}(E_\psi)$ is a single point. This guarantees that μ has degree 1.

Proof. We can see that

$$\mathfrak{p}(\Lambda) = \check{\mathfrak{g}}^{\mathrm{Ad}(E_\psi)} + \check{\mathfrak{g}}^{\mathrm{Ad}(E_\psi)^2}\cap\mathrm{Ad}(E_\psi)\check{\mathfrak{g}} + \check{\mathfrak{g}}^{\mathrm{Ad}(E_\psi)^3}\cap\mathrm{Ad}(E_\psi)^2\check{\mathfrak{g}} + \cdots,$$

So the centralizer $\check{G}(\Lambda)^{E_\psi} \subseteq P(\Lambda)$, which means that the fiber $\mu^{-1}(E_\psi)$ is a single point. The operator E_ψ carry the sum of non-negative eigenspaces of $\mathrm{ad}(\lambda)$ onto the sum of positive eigenspaces, i.e. $\mathrm{Ad}(E_\psi)\mathfrak{p}(\Lambda) = \mathfrak{n}(\Lambda)$. In other words, the differential of the map:

$$P(\Lambda) \to \mathfrak{n}(\lambda), \quad p \mapsto \mathrm{Ad}(p)E_\psi,$$

is surjective at the identity coset, so the orbit $\check{G}(\Lambda)\cdot E_\psi$ is dense in $\mathfrak{n}(\lambda)$, which means that E_ψ is a regular element in $\mathcal{N}_{\mathcal{P}(\Lambda)}$. ∎

Definition 20.3. *An Arthur parameter* ψ *is called* unipotent *if the restriction to* $\mathbb{C}^* \subseteq W_{\mathbb{R}}$ *is trivial.*

Remark 20.4. *In this case,* $\psi : \mathrm{SL}_2(\mathbb{C}) \times \Gamma \to \check{G}^{\Gamma}$. *So* $\lambda = \lambda_1$

Theorem 20.5. *Fix an Arthur paramter* $\psi : \Gamma \times \mathrm{SL}_2(\mathbb{C}) \to \check{G}^{\Gamma}$, *let* $\lambda = d\psi \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}$, $\mathcal{O} = \check{G} \cdot \lambda \subseteq \check{\mathfrak{g}}$ *the corresponding semisimple orbits. Suppose that* $\phi_\psi = (y, \lambda)$, *then we have an open and closed subvariety of the geometric parameter space* $X_y(\mathcal{O}, \check{G}^{\Gamma}) \simeq \check{G} \times^{K(y)} (\check{G}(\Lambda)_0/P(\Lambda))$.

Then there are bijections between:

- *equivalence classes of unipotent Arthur parameters supported on* $X_y(\mathcal{O}, \check{G}^{\Gamma})$;
- *regular* $K(y)$*-orbits in* $\check{G}(\Lambda)_0/P(\Lambda)$;
- *regular* $K(y)$*-orbits in* $\mathfrak{s}(y) \cap \mathcal{N}_{\mathcal{P}(\Lambda)}$.

Proof. Given a unipotent Arthur parameter ψ, we can associate to it a nilpotent element $E_\psi \in \mathfrak{s}(y) \cap \mathcal{N}_{\mathcal{P}(\Lambda)}$ as above, then $K(y) \cdot E_\psi$ is a regular orbit in $\mathfrak{s}(y) \cap \mathcal{N}_{\mathcal{P}(\Lambda)}$.

Conversely, given a regular $K(y)$-orbit in $\mathfrak{s}(y) \cap \mathcal{N}_{\mathcal{P}(\Lambda)}$, let e be a representative. We can construct a homomorphism $\psi_1' : \mathrm{SL}_2(\mathbb{C}) \to \check{G}$ such that $d\psi_1' \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = e$, and

$$\psi_1'(\theta_{\mathrm{SL}_2}(x)) = \theta_y(\psi_1'(x)).$$

Define $y_0 = y\psi_1' \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}$

Then (y_0, ψ_1') is the corresponding unipotent Arthur parameter. □

Remark 20.6. *In other words, this theorem tell us the following diagram:*

$$\begin{tikzcd}
\boxed{\check{G}\text{-orbits in } \check{G} \times^{K(y)} (\check{G}(\Lambda)_0/P(\Lambda))} \arrow[r, leftrightarrow, "1-1"] & \boxed{K(y)\text{-orbits in } \check{G}(\Lambda)_0/P(\Lambda)} \\
\boxed{\text{unipotent } A\text{-parameters supported on } X_y(\mathcal{O}, \check{G}^{\Gamma})} \arrow[r, leftrightarrow, "1-1"] \arrow[u, hook] & \boxed{\text{regular } K(y)\text{-orbits in } \check{G}(\Lambda)_0/P(\Lambda)} \arrow[u, hook]
\end{tikzcd}$$

Theorem 20.7. *Under the above setting, fix a unipotent Arthur parameter* ψ *supported on* $X_y(\mathcal{O}, \check{G}^{\Gamma})$, *let* S *be the corresopnding* $\check{G}$*-orbit in* $X_y(\mathcal{O}, \check{G}^{\Gamma})$, S_K *be the corresponding* $K(y)$*-orbit in* $\check{G}(\Lambda)_0/P(\Lambda)$, *and* $\mathcal{O}_K$ *be the corresponding* $K(y)$*-orbit in* $\mathfrak{s}(y) \cap \mathcal{N}_{\mathcal{P}(\Lambda)}$.

Then there are bijections between:

- *irrducible* $(\check{\mathfrak{g}}(\Lambda), K(y)^{alg})$*-modules* M *annihilated by* $I_{\mathcal{P}(\Lambda)}$ *with* $\overline{\mathcal{O}_K} \subseteq \mathrm{AV}(M)$;
- *irreducible* $K(y)^{alg}$*-equivariant* $\mathcal{D}$*-modules* $\mathcal{M}$ *on* $\check{G}(\Lambda)_0/P(\Lambda)$ *with* $T^*_{S_K}(\check{G}(\Lambda)_0/P(\Lambda)) \subseteq \mathrm{Ch}(\mathcal{M})$;
- *irreducible* $\check{G}$*-equivariant* $\mathcal{D}$*-modules* $\mathcal{N}$ *on* $X_y(\mathcal{O}, \check{G}^{\Gamma})$ *with* $T^*_S(X_y(\mathcal{O}, \check{G}^{\Gamma})) \subseteq \mathrm{Ch}(\mathcal{N})$;
- *Arthur packets* $\Pi^{Art}_\psi(G/\mathbb{R})$ *associated to* ψ.

Corollary 20.8. *Fix a nilpotent orbit* $\check{\mathcal{O}} \subseteq \check{\mathfrak{g}}$, *attached to it is a semisimple orbit* $\check{\mathcal{O}}_{ss} \subseteq \check{\mathfrak{g}}$ *then*

$$\mathrm{Unip}_{\check{\mathcal{O}}}(G/\mathbb{R}) = \bigcup_{\psi \text{ unipotent}, \mathcal{O}_\psi = \check{\mathcal{O}}_{ss}} \Pi^{Art}_\psi(G/\mathbb{R}).$$

Remark 20.9.

$$\mathrm{Unip}_{\check{\mathcal{O}}}(G/\mathbb{R}) = \{\textit{irreducible representations in } \mathrm{Irr}_{\check{\mathcal{O}}_{ss}}(G/\mathbb{R}) \textit{ with minimal GK-dim}\}$$

Example 20.10. *$G = G_1 \times G_1$ where G_1 is a complex connected reductive algebraic group viewed as a real group. Then $\check{G}^\Gamma = (\check{G}_1 \times \check{G}_1) \rtimes \{1, \delta\}$. A homomorphism $\psi : \mathrm{SL}_2(\mathbb{C}) \to \check{G}$ is determined by two homomorphisms $\psi_1, \psi_2 : \mathrm{SL}_2(\mathbb{C}) \to \check{G}_1$, we can check that ψ comes from a unipotent Arthur parameter if and only if ψ_1 and ψ_2 are conjugate to each other. So unipotent Arthur parameters of G are in bijection with homomorphisms $\psi_1 : \mathrm{SL}_2(\mathbb{C}) \to \check{G}_1$. Therefore, equivalence classes of unipotent Arthur parameters of G_1 are in bijection with nilpotent orbits in $\check{\mathfrak{g}}_1$.*

21. 2025.11.4

Some questions about unitary dual problem

- How can we reduce the classification of irreducible unitary $(\mathfrak{g}, K)$-module to the classification of such module with real infinitesimal character?
- Why there always exist a $\mathfrak{u}_{\mathbb{R}}$-invariant Hermitian form on irreducible $(\mathfrak{g}, K)$-module with real infinitesimal character?
- what is Jantzen filtration of a standard $(\mathfrak{g}, K)$-module? Why it coincides with Hodge filtration?

Let G be a connected reductive group over $\mathbb{C}$, $\mathcal{B}$ the variety of all Borel subalgebras of $\mathfrak{g}$, ${}^a\mathfrak{h}$ is the universal Cartan subalgebra. For any Borel subalgebra $\mathfrak{b} \in \mathcal{B}$, we have a canonical isomorphism:

$$\phi_{\mathfrak{b}} : {}^a\mathfrak{h} \simeq \mathfrak{h},$$

such that negative roots in ${}^a\mathfrak{h}^*$ correspond to roots in $\Delta(\mathfrak{h}, \mathfrak{n})$.

We can define a G-equivariant vector bundle $\mathcal{L}_\lambda$ over $\mathcal{B}$ for any $\lambda \in {}^a\mathfrak{h}^*$ by:

$$\mathcal{L}_\lambda|_{\mathfrak{b}} = \mathbb{C}_\lambda := \text{1-dimensional representation of } B \text{ defined by } \lambda \circ \phi_{\mathfrak{b}}^{-1}.$$

$$\boxed{\text{Weight lattice } \Lambda \subseteq {}^a h^*} \xleftrightarrow{1-1} \boxed{G\text{-equivariant line bundles on } \mathcal{B}}$$

Under this correspondence, $\mathcal{L}_\lambda$ is ample if and only if λ is regular dominant. $\mathcal{L}_{-2\rho}$ is the canonical line bundle $\wedge^{top} T^*\mathcal{B}$ on $\mathcal{B}$.

22. 2025.11.19

22.1. compute annihilator variety.

$$\boxed{\mathrm{Prim}_\nu(\mathrm{U}(\mathfrak{g}))} \xleftrightarrow{1-1} \boxed{\text{left cells in } \mathrm{Coh}_\nu(\mathfrak{g},\mathfrak{b})} \longrightarrow \boxed{\text{double cells in } \mathrm{Coh}_\nu(\mathfrak{g},\mathfrak{b})}$$

$$I \in \mathrm{Prim}(\mathrm{U}(\mathfrak{g})) \longmapsto \mathcal{C}^L_{\nu,I} \longmapsto \mathcal{C}^{LR}_{\nu,I}$$

$\mathrm{span}(\mathcal{C}^{LR}_{\nu,I})$ is a representation of the integral Weyl group $W(\nu)$, all irreducible constituents of $\mathrm{span}(\mathcal{C}^{LR}_{\nu,I})$ lies in a single double cell, and contains a unique special representation of $W(\nu)$

$$\sigma_{\nu,I}.$$

Then by results of Joseph, we have:

$$\mathrm{AV}(I) = \mathcal{O}(j^W_{W(\nu)}(\sigma_{\nu,I})).$$

Which means that to compute the annihilator variety of $\mathrm{L}(w\nu)$, we only need to compute the corresponding cell representation.

In the case of $(\mathfrak{g}, K)$-modules, the method is the same:

$$\begin{array}{c} X\text{: Irreducible } (\mathfrak{g},K)\text{-module} \\ \rightsquigarrow \\ C\text{: Harish-Chandra cell containing } X \\ \rightsquigarrow \\ \sigma_C\text{: special representation of } W(\nu) \text{ contained in the cell representation} \\ \rightsquigarrow \\ \mathcal{O}(j^W_{W(\nu)}(\sigma_C)) = \mathrm{AV}(X) \end{array}$$

Remark 22.1. *The key point is the result of Joseph* [Jos85] *the Springer correspondence of cell representation gives the annihilator variety. However, I'm not fully understand the details.*

22.2. annihilator varieties are special.

j-induction of parabolic subgroups preserves special representations.

To proove that the annihilator variety of irreducible $(\mathfrak{g}, K)$-module is closure of a single special nilpotent orbit, only need to check the case when infinitesimal character has good pariety and bad pariety.

For example, in the case of $G = \mathrm{Sp}_{2n}(\mathbb{R})$, we have:

$$\mathrm{Coh}_{\Lambda_b}(\mathrm{SO}(n_b,n_b)) \otimes \mathrm{Coh}_{\Lambda_g}(\mathrm{Sp}_{2n_g}(\mathbb{R})) \simeq \mathrm{Coh}_{\Lambda}(\mathrm{Sp}_{2\mathrm{n}}(\mathbb{R})).$$

We need to show that the j-induction of any special repn appears in $\mathrm{Coh}_{\Lambda_b}(\mathrm{SO}(n_b,n_b))$ and any special repn appears in $\mathrm{Coh}_{\Lambda_g}(\mathrm{Sp}_{2n_g}(\mathbb{R}))$ gives a special repn in $\mathrm{Coh}_{\Lambda}(\mathrm{Sp}_{2\mathrm{n}}(\mathbb{R}))$.

- Problem: How to compute the j-induction explicitly?
- Problem: the special repn of each cell always appears?

23. 2025.11.21

Some problems/plans

- Use counting methods to prove that all annihilator variety of irreducible $(\mathfrak{g}, K)$-modules of orthogonal or special groups are special.
- Use counting methods to prove that all annihilator variety of irreducible genuine $(\mathfrak{g}, K)$-modules of metaplectic groups are metaplectic special.
- By computation, prove that in the case of metaplectic groups, metaplectic quasi-admissible equivalent to metaplectic special. solved! Yes
- Prove that for metaplectic groups, whittaker support of irreducible genuine $(\mathfrak{g}, K)$-modules are metaplectic quasi-admissible. Use arguments of Gomez-Gourevitch-Sahi
- Prove associated cycles coincides with whittaker cycles for irreducible $(\mathfrak{g}, K)$-modules of real classical groups and metaplectic groups.

24. 2025.11.24

For a Weyl group W, elements of W are participated into double cells, left cells, etc. Each double cell $\mathcal{C}^{LR}$ contains a unique special representation $\sigma_{\mathcal{C}^{LR}}$ of W. And each left cell $\mathcal{C}^{L}$ corresponds to a primitive ideal $I_{\mathcal{C}^L}$ of $\mathrm{U}(\mathfrak{g})$ with infinitesimal character ρ. For each element $w \in W$, there are two processes to get a nilpotent orbit $\mathcal{O}_w$ in $\mathfrak{g}^*$:

$$w \longmapsto \mathcal{C}_w^{LR} \longmapsto \sigma_{\mathcal{C}_w^{LR}} \longmapsto \mathcal{O}(\sigma_{\mathcal{C}_w^{LR}});$$

$$w \xmapsto{RS} P(w) \longmapsto \mathcal{O}_{P(w)} \longmapsto \mathcal{O}_{P(w),s}.$$

The question is: are these two nilpotent orbits the same? i.e. $\mathcal{O}(\sigma_{\mathcal{C}_w^{LR}}) = \mathcal{O}_{P(w),s}$?
answer: Yes! [BV82]

25. 2026.3.3

G is a linear real reductive group, $\widetilde{G}$ is a double cover of G.

Theorem 25.1. *$\widetilde{G}_0$ is a linear group (i.e. there is an injective homomorphism $\widetilde{G}_0 \to \mathrm{GL}_n(\mathbb{R})$ for some n) equivalent to the existence of a finite-dimensional genuine representation of $\widetilde{G}$.*

Proof. If $\widetilde{G}_0$ is a linear group, there are two possibilities:

- $\varepsilon \notin \widetilde{G}_0$: then $\{1, \varepsilon\} \times \widetilde{G}_0$ is a subgroup of $\widetilde{G}$ with finite index, and it has a finite-dimensional representation such that ε acts by -1, so we get a finite-dimensional genuine representation of $\widetilde{G}$ by induction.
- $\varepsilon \in \widetilde{G}_0$: denote $\widetilde{G}_0$ by H, then $H = A \times {}^0H$ as a Nash group, and ${}^0H$ is also a linear group. So there is a closed embedding of Nash groups ${}^0H \hookrightarrow \mathrm{GL}_n(\mathbb{R})$ for some n (${}^0H$ has compact center), take Zariski closure of ${}^0H$ in $\mathrm{GL}_n(\mathbb{C})$ we get the complexification ${}^0H_{\mathbb{C}}$ of ${}^0H$, there is an algebraic matrix coefficient of ${}^0H_{\mathbb{C}}$ such that ε acts on it by -1, and it generates a finite-dimensional genuine representation of $\widetilde{G}$ by induction.

Conversely, if there is a finite-dimensional genuine representation of $\widetilde{G}$, we also have two possibilities:

- $\varepsilon \notin \widetilde{G}_0$: then $\widetilde{G}_0 \to G$ is injective, since G is a linear group, $\widetilde{G}_0$ is also a linear group.
- $\varepsilon \in \widetilde{G}_0$: then we can find a finite-dimensional representation E_1 of $\widetilde{G}_0$ such that ε acts by -1. We also have a finite-dimensional faithful representation E_2 of G, we can regard it as a representation of $\widetilde{G}_0$ by pullback, and ε acts trivially on E_2. So $E_1 \oplus E_2$ is a finite-dimensional faithful representation of $\widetilde{G}_0$.

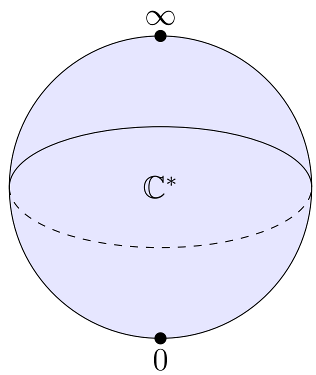
□

Remark 25.2. *After I have a sleep, I find that $\widetilde{G}_0$ can be repleced by $\widetilde{G}$ in this theorem, and the proof is even easier.*

26. 2026.3.21

In this section, we illustrate the geometry of the flag variety of $\mathrm{SL}_2(\mathbb{C})$, which serve as a simple example for understanding more general phenomena.

$K = \mathbb{C}^*$ action has three orbits on $\mathbb{P}^1(\mathbb{C})$: two closed orbits $\{0\}$ and $\{\infty\}$, and one open orbit $\mathbb{C}^*$. The conormal bundle of the open orbit is the zero section, while the conormal bundle of the closed orbits are the cotangent fiber at 0 and ∞ respectively. So we have three conormal bundles, which are all Lagrangian subvarieties in $T^*\mathbb{P}^1(\mathbb{C})$.

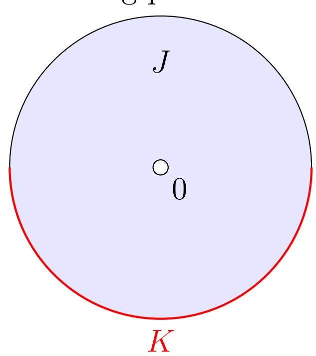


For a K-equivariant perverse sheaf $\mathcal{F}^\bullet$ on it, we calculate its characteristic cycle $\mathrm{CC}(\mathcal{F}^\bullet)$ use the formula

$$\chi_S^{mic} = \sum_{S' > S} c(S, S')\chi_{S'}^{loc},$$

only need to determine the coefficients $c(S, S')(= \sum_i (-1)^i H_c^i(J \cap S', K \cap S'; \mathbb{C}))$ for $S' > S$, where (J, K) is the normal Morse datum of S, which is well-defined up to homeomorphism.

- For the open orbit $\mathbb{C}^*$, we have $S' = \mathbb{C}^*$ and $c(\mathbb{C}^*, \mathbb{C}^*) = -1$.
- For the closed orbit $\{0\}$, we have $S' = \{0\}, \mathbb{C}^*$, and $c(\{0\}, \{0\}) = 1$, for $c(\{0\}, \mathbb{C}^*)$, the pair $(J \cap S', K \cap S')$ is the following picture



therefore $c(\{0\}, \mathbb{C}^*) = \sum_i (-1)^i H_c^i(J \cap \mathbb{C}^*, K \cap \mathbb{C}^*; \mathbb{C}) =^? -1$.

So we have

$$\begin{aligned}
\chi_0^{mic} &= \chi_0^{loc} - \chi_{\mathbb{C}^*}^{loc}, \\
\chi_\infty^{mic} &= \chi_\infty^{loc} - \chi_{\mathbb{C}^*}^{loc}, \\
\chi_{\mathbb{C}^*}^{mic} &= -\chi_{\mathbb{C}^*}^{loc}.
\end{aligned}$$

27. 2026.3.22

Construuctible sheaves on $\mathbb{C} = \mathbb{C}^* \sqcup \{0\}$.

Denote $j : \mathbb{C}^* \hookrightarrow \mathbb{C}$ the open inclusion, $i : \{0\} \hookrightarrow \mathbb{C}$ the closed inclusion. Then we have the following six functors:

$$\begin{tikzcd} D^b(\mathbb{C}^*) \arrow[r, shift left=2, "j_!"] \arrow[r, shift right=2, "j_*"'] & D^b_c(\mathbb{C}) \arrow[l, "j^*=j^!" description] \arrow[r, shift left=2, "i^*"] \arrow[r, shift right=2, "i^!"'] & D^b(\{0\}) \arrow[l, "i_*=i_!" description] \end{tikzcd}$$

$\mathcal{L}$ is a local system on $\mathbb{C}^*$.

$j_!\mathcal{L}$ has no quotient supported on $\{0\}$, j_* has no subobject supported on $\{0\}$. $j_!\mathcal{L}[1]$ and $j_*\mathcal{L}[1]$ are perverse sheaves on $\mathbb{C}$.

By easy calculation of group cohomology, we have:

$$H^i(j_*\mathcal{L}[1]_0) = \begin{cases} 0 & i \neq 0, -1 \\ V_\mu & i = 0 \\ V^\mu & i = -1 \end{cases}$$

References

[BB82] W. Borho and J.-L. Brylinski, Differential operators on homogeneous spaces. I. Irreducibility of the associated variety for annihilators of induced modules, Invent. Math. **69** (1982), no. 3, 437–476; MR0679767

[BV80] D. M. Barbasch and D. A. Vogan Jr, The local structure of characters, J. Functional Analysis **37** (1980), no. 1, 27–55; MR0576644

[BV82] D. M. Barbasch and D. A. Vogan Jr, Primitive ideals and orbital integrals in complex classical groups, Math. Ann. **259** (1982), no. 2, 153–199; MR0656661

[GGS21] R. Gomez Muñoz, D. Gourevitch and S. Sahi, Whittaker supports for representations of reductive groups, Ann. Inst. Fourier (Grenoble) **71** (2021), no. 1, 239–286; MR4275869

[GZ14] R. Gomez Muñoz and C.-B. Zhu, Local theta lifting of generalized Whittaker models associated to nilpotent orbits, Geom. Funct. Anal. **24** (2014), no. 3, 796–853; MR3213830

[HTT08] R. Hotta, K. Takeuchi and T. Tanisaki, *D-modules, perverse sheaves, and representation theory*, translated from the 1995 Japanese edition by Takeuchi, Progress in Mathematics, 236, Birkhäuser Boston, Boston, MA, 2008; MR2357361

[Jos85] A. Joseph, On the associated variety of a primitive ideal, J. Algebra **93** (1985), no. 2, 509–523; MR0786766

[KL80] D. A. Kazhdan and G. Lusztig, A topological approach to Springer's representations, Adv. in Math. **38** (1980), no. 2, 222–228; MR0597198

[KT84] M. Kashiwara and T. Tanisaki, The characteristic cycles of holonomic systems on a flag manifold related to the Weyl group algebra, Invent. Math. **77** (1984), no. 1, 185–198; MR0751138

[KV95] A. W. Knapp and D. A. Vogan Jr, *Cohomological induction and unitary representations*, Princeton Mathematical Series, 45, Princeton Univ. Press, Princeton, NJ, 1995; MR1330919

[MW87] C. Mœglin and J.-L. Waldspurger, Modèles de Whittaker dégénérés pour des groupes p-adiques, Math. Z. **196** (1987), no. 3, 427–452; MR0913667

[Nev99] M. Nevins, Admissible nilpotent coadjoint orbits of p-adic reductive Lie groups, Represent. Theory **3** (1999), 105–126; MR1698202

[Sch87] J. O. Schwartz, *THE DETERMINATION OF THE ADMISSIBLE NILPOTENT ORBITS IN REAL CLASSICAL GROUPS*, ProQuest LLC, Ann Arbor, MI, 1987; MR2941135

[SGA3] M. Demazure, A. Grothendieck. *Schémas en Groupes (SGA3) I, II, III*, Lecture Notes in Mathematics 151, 152, 153, Springer, New York (1970).

[Sun15] B. Sun, Almost linear Nash groups, Chinese Ann. Math. Ser. B **36** (2015), no. 3, 355–400; MR3341164